H3K79me2 association with gene age and expression dynamics

Here, I load the supplementary table from Jeromine's hPTM signatures and examine the association between the H3K79me2 and the distribution of gene expression.

```
library(tidyverse)
library(ggrastr)
library(readxl)
library(scales)
```

```
readxl::excel_sheets("data/dataset_S3_master_tables.xlsx")
```

```
## [1] "Ectocarpus"              "Desmarestia-dudresnayi" "Desmarestia-herbacea"
## [4] "Undaria"                 "Scitosyphon"            "Schizocladia"
```

```
Ec_k79 <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Ectocarpus") |>
        dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("peak_K79"), rank)
Ddud_k79 <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Desmarestia-dudresnayi") |>
        dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("peak_K79"), rank)
Dher_k79 <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Desmarestia-herbacea") |>
        dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("peak_K79"), rank)
Und_k79 <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Undaria") |>
        dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("peak_K79"), rank)
Spro_k79 <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Scitosyphon") |>
        dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("peak_K79"), rank)
Si_k79 <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Schizocladia") |>
        dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("peak_K79"), rank)
```

Note: Scytosiphon is misspelt in the table

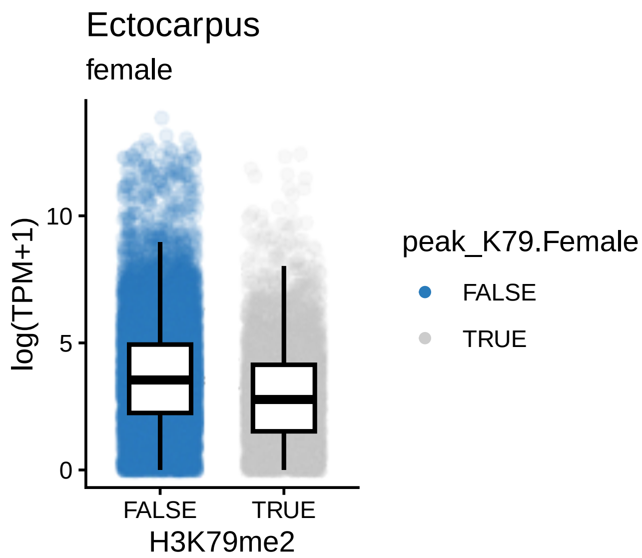
```
preprocess_ps_sig <- function(Species_ps){
        Species_ps.filt <- Species_ps |>
                tidyr::drop_na() |>
                dplyr::mutate(rank = as.numeric(rank)) |>
                tidyr::drop_na(rank)
        if(any(colnames(Species_ps.filt) %in% c("peak_K79.Female", "peak_K79.Male"))){
                Species_ps.filt2 <-
                        Species_ps.filt |>
                        dplyr::mutate(Signature.Female = factor(
                                Signature.Female, levels = paste0('S',1:16))) |>
                        dplyr::mutate(Signature.Male = factor(
                                Signature.Male, levels = paste0('S',1:16)))
        }else{
                Species_ps.filt2 <-
                        Species_ps.filt |>
                        dplyr::mutate(Signature = factor(
                                Signature, levels = paste0('S',1:16)))
        }
        return(Species_ps.filt2)
}
```

```r
color_table <-
        tibble::tibble(
                Signature = paste0('S',1:16),
                Colour = c("#a31d21", "#e43a30", "#f36f57", "#f8a68d", "#fddfd1", "#1b733b",
            "#2daf66", "#6ec283", "#aed9b1", "#9392c8", "#bfbfdf", "#e9eef8",
            "#2a74bb", "#61abd7", "#afd1e5", "#999999")
        )

plot_style <- function(p_in){
        p_out <-
                p_in +
                ggrastr::rasterise(ggplot2::geom_jitter(width = 0.3, alpha = 0.1)) +
                ggplot2::geom_boxplot(width = 0.5, size = 0.8, outliers = FALSE, colour = "black") +
                ggplot2::theme_classic() +
                ggplot2::scale_colour_manual(values = c("#297BBB", "grey80")) +
                # ggplot2::theme(legend.position = "none") +
                ggplot2::guides(colour = guide_legend(override.aes = list(alpha = 1))) +
                # ggplot2::coord_flip() +
                ggplot2::scale_y_continuous(breaks = scales::breaks_pretty())
        return(p_out)
}

plot_style2 <- function(p_in){
        p_out <-
                p_in +
                ggrastr::rasterise(ggplot2::geom_jitter(width = 0.3, alpha = 0.1)) +
                ggplot2::geom_boxplot(width = 0.5, size = 0.8, outliers = FALSE, colour = "black") +
                ggplot2::theme_classic() +
                ggplot2::scale_colour_manual(values = c("#297BBB", "grey80")) +
                # ggplot2::theme(legend.position = "none") +
                ggplot2::guides(colour = guide_legend(override.aes = list(alpha = 1))) +
                # ggplot2::coord_flip() +
                ggplot2::scale_y_continuous(breaks = scales::breaks_pretty())
        return(p_out)
}
```

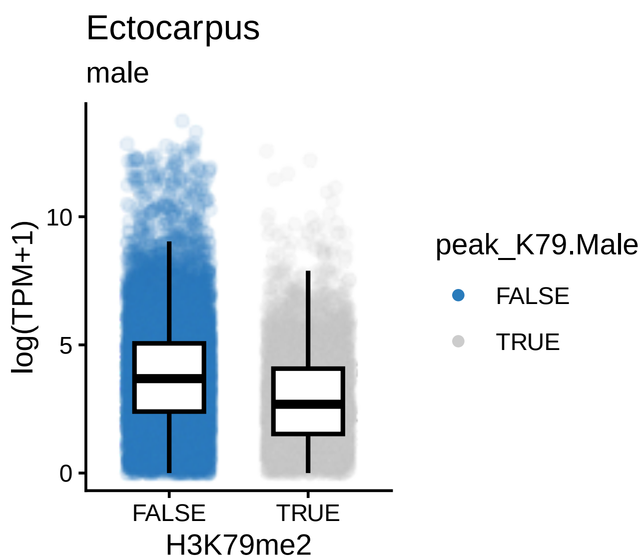
H3K79me2 and mean expression or gene age.

Ectocarpus

```r
Ec_k79.filt <-
        Ec_k79 |>
        tidyr::drop_na()

# expression
Ec_k79.filt |>
        ggplot2::ggplot(aes(x = peak_K79.Female, colour = peak_K79.Female, y = log2_TPM_plus_1.Female))
        plot_style() +
        ggplot2::labs(x = "H3K79me2", y = "log(TPM+1)",
                      title = "Ectocarpus", subtitle = "female")
```

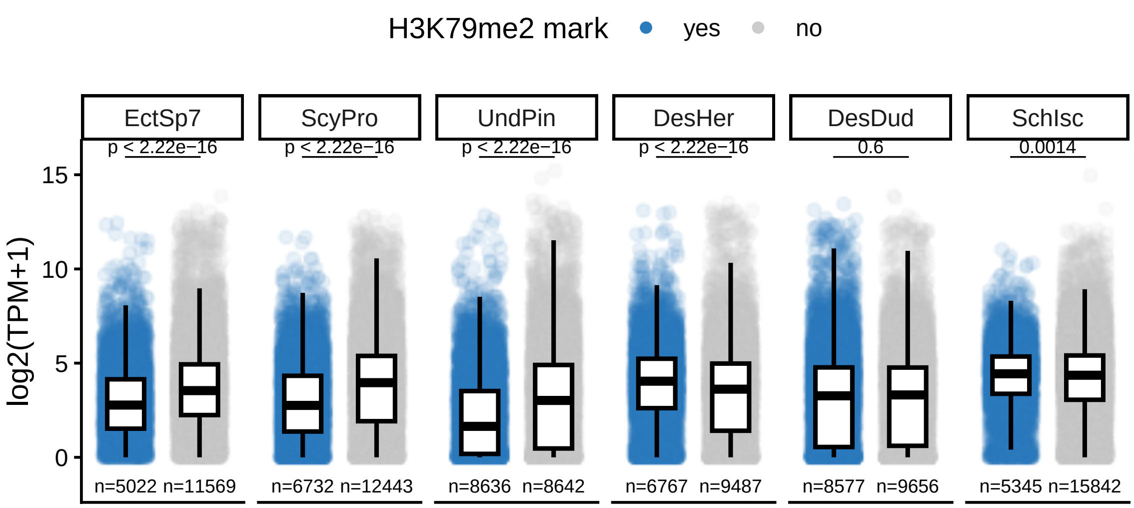



```
Ec_k79.filt |>
        ggplot2::ggplot(aes(x = peak_K79.Male, colour = peak_K79.Male, y = log2_TPM_plus_1.Male)) |>
        plot_style() +
        ggplot2::labs(x = "H3K79me2", y = "log(TPM+1)",
                      title = "Ectocarpus", subtitle = "male")
```



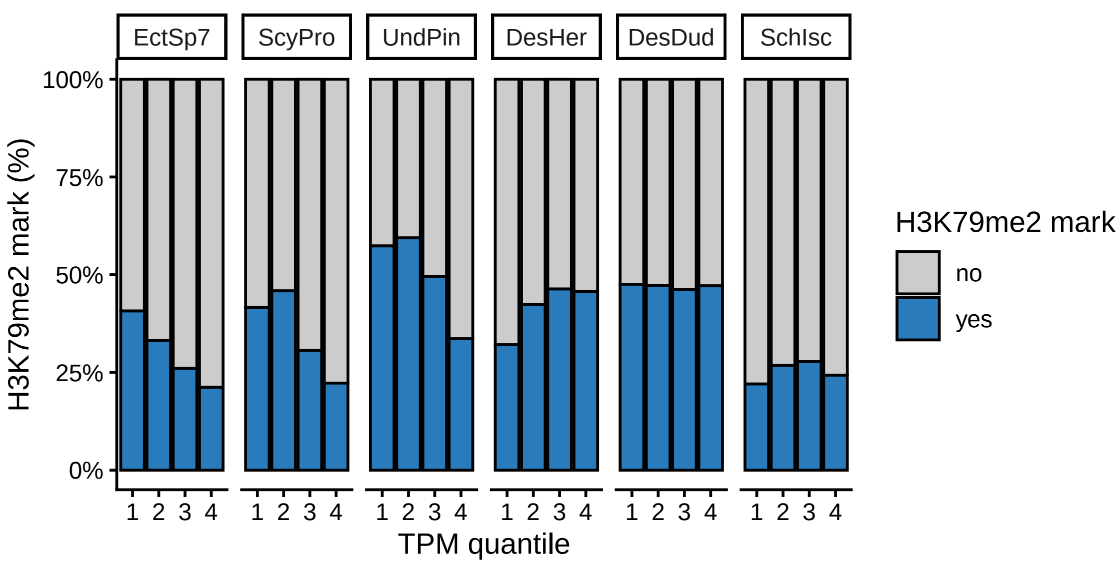

```
all_k79.filt <-
        Ec_k79 |> dplyr::mutate(Sp = "EctSp7") |>
        rbind(Spro_k79 |> dplyr::mutate(Sp = "ScyPro")) |>
        rbind(Und_k79 |> dplyr::mutate(Sp = "UndPin")) |>
        rbind(Dher_k79 |> dplyr::mutate(Sp = "DesHer")) |>
        dplyr::select(gene_id,
                      log2_TPM_plus_1 = log2_TPM_plus_1.Female,
                      peak_K79 = peak_K79.Female,
                      Sp) |>
        rbind(Ddud_k79 |> dplyr::mutate(Sp = "DesDud") |> dplyr::select(1,3,4,6)) |>
        rbind(Si_k79 |> dplyr::mutate(Sp = "SchIsc") |> dplyr::select(1,3,4,6)) |>
        tidyr::drop_na()
```

```r
all_k79.filt |>
        dplyr::mutate(Sp = factor(Sp, levels = c(
                "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
        ))) |>
        dplyr::mutate(peak_K79 = case_when(
                peak_K79 == TRUE ~ "yes",
                .default = "no"
        )) |>
        dplyr::mutate(peak_K79 = factor(peak_K79, levels = c("yes", "no"))) |>
        ggplot2::ggplot(aes(x = peak_K79, colour = peak_K79, y = log2_TPM_plus_1)) |>
        plot_style() +
        ggplot2::facet_wrap(.~Sp, nrow = 1) +
        ggplot2::labs(x = "", y = "log2(TPM+1)", colour = "H3K79me2 mark") +
        ggplot2::theme(axis.text.x = element_blank(), axis.ticks.x = element_blank(),
                       legend.position = "top") +
        ggpubr::stat_compare_means(
                label = "p.adj",
                method = "wilcox.test",
                comparisons = list(c("yes", "no")),
                p.adjust.method = "bonferroni",
                tip.length = 0,
                size = 2.5
        ) +
        EnvStats::stat_n_text(size = 2.5)
```




```r
all_k79.filt.TPM <-
        Ec_k79 |> dplyr::mutate(Sp = "EctSp7") |>
        rbind(Spro_k79 |> dplyr::mutate(Sp = "ScyPro")) |>
        rbind(Und_k79 |> dplyr::mutate(Sp = "UndPin")) |>
        rbind(Dher_k79 |> dplyr::mutate(Sp = "DesHer")) |>
        dplyr::select(gene_id,
                      TPM = TPM.Female,
                      peak_K79 = peak_K79.Female,
                      Sp) |>
        rbind(Ddud_k79 |> dplyr::mutate(Sp = "DesDud") |> dplyr::select(1,2,4,6)) |>
        rbind(Si_k79 |> dplyr::mutate(Sp = "SchIsc") |> dplyr::select(1,2,4,6)) |>
        tidyr::drop_na()
```

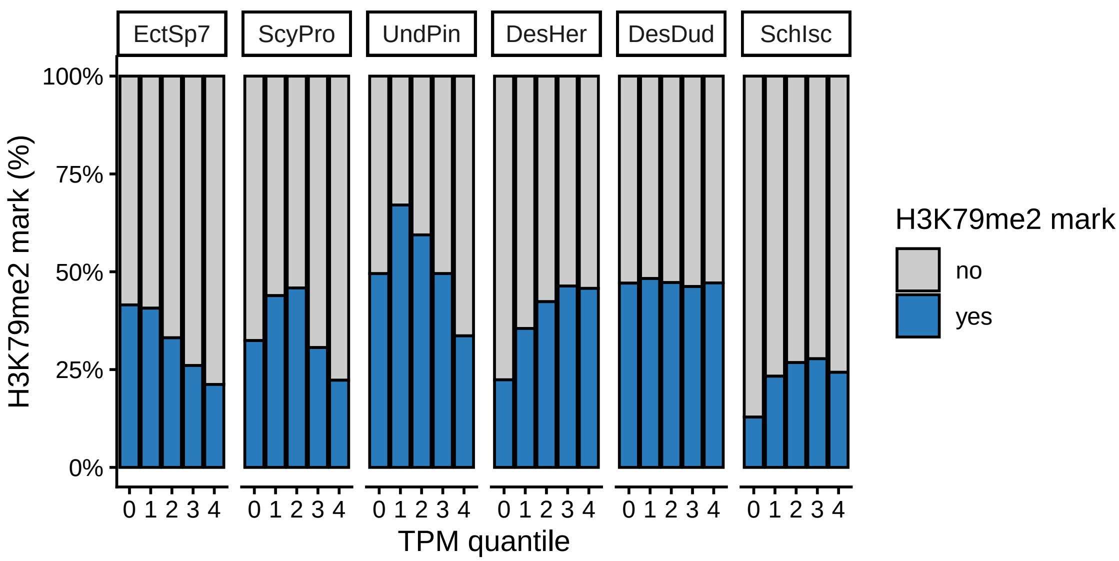
```r
all_k79.filt.TPM.filt <-
        all_k79.filt.TPM |>
        dplyr::mutate(TPM_quantile = ntile(TPM, 4), .by = Sp)
        # dplyr::mutate(TPM_quantile = dplyr::case_when(
        #          TPM == 0 ~ 0,
        #          TPM > 0 ~ ntile(TPM, 4)
        # ),
        # .by = Sp)

all_k79.filt.TPM.filt |>
        dplyr::mutate(Sp = factor(Sp, levels = c(
                "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
        ))) |>
        dplyr::mutate(peak_K79 = case_when(
                peak_K79 == TRUE ~ "yes",
                .default = "no"
        )) |>
        # dplyr::mutate(peak_K79 = factor(peak_K79, levels = c("yes", "no"))) |>
        ggplot2::ggplot(aes(x = factor(TPM_quantile), fill = peak_K79)) +
        ggplot2::geom_bar(position = "fill", colour = "black") +
        ggplot2::scale_y_continuous(labels = scales::percent_format()) +
        ggplot2::labs(
                x = "TPM quantile",
                y = "H3K79me2 mark (%)",
                fill = "H3K79me2 mark"
                ) +
        ggplot2::theme_classic() +
        ggplot2::scale_fill_manual(values = c("grey80", "#297BBB")) +
        ggplot2::facet_wrap(.~Sp, nrow = 1)
```



Plot the quantiles including "not expressed" category with TPM == 0

```r
all_k79.filt.TPM.filt2 <-
        all_k79.filt.TPM |>
        dplyr::mutate(TPM_quantile = dplyr::case_when(
                TPM == 0 ~ "0",
```

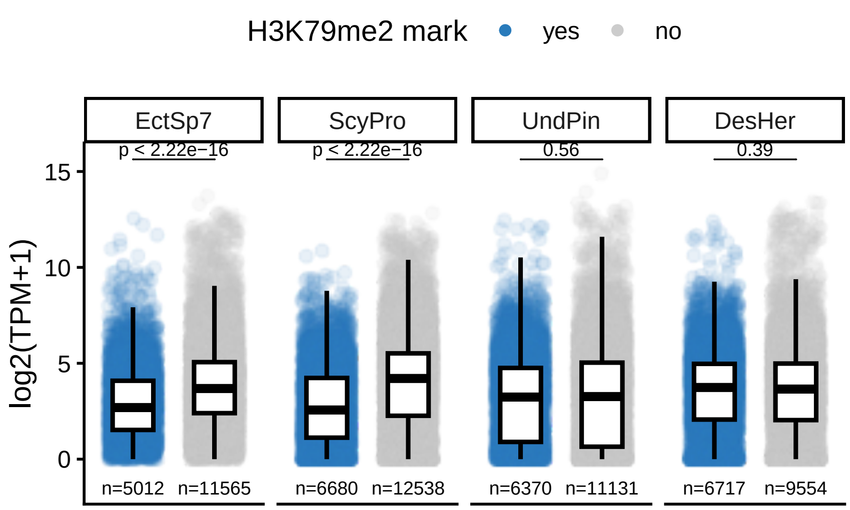
```
                TPM > 0 ~ as.character(ntile(TPM, 4))
        ),
        .by = Sp)
all_k79.filt.TPM.filt2 |>
        dplyr::mutate(Sp = factor(Sp, levels = c(
                "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
        ))) |>
        dplyr::mutate(peak_K79 = case_when(
                peak_K79 == TRUE ~ "yes",
                .default = "no"
        )) |>
        # dplyr::mutate(peak_K79 = factor(peak_K79, levels = c("yes", "no"))) |>
        ggplot2::ggplot(aes(x = factor(TPM_quantile, levels = c("0","1","2","3","4")), fill = peak_K79))
        ggplot2::geom_bar(position = "fill", colour = "black") +
        ggplot2::scale_y_continuous(labels = scales::percent_format()) +
        ggplot2::labs(
                x = "TPM quantile",
                y = "H3K79me2 mark (%)",
                fill = "H3K79me2 mark"
        ) +
        ggplot2::theme_classic() +
        ggplot2::scale_fill_manual(values = c("grey80", "#297BBB")) +
        ggplot2::facet_wrap(.~Sp, nrow = 1)
```




```
all_k79.filt.M <-
        Ec_k79 |> dplyr::mutate(Sp = "EctSp7") |>
        rbind(Spro_k79 |> dplyr::mutate(Sp = "ScyPro")) |>
        rbind(Und_k79 |> dplyr::mutate(Sp = "UndPin")) |>
        rbind(Dher_k79 |> dplyr::mutate(Sp = "DesHer")) |>
        dplyr::select(gene_id,
                      log2_TPM_plus_1 = log2_TPM_plus_1.Male,
                      peak_K79 = peak_K79.Male,
                      Sp) |>
        # rbind(Ddud_k79 |> dplyr::mutate(Sp = "DesDud") |> dplyr::select(1,3,4,6)) |>
```

```
        # rbind(Si_k79 |> dplyr::mutate(Sp = "SchIsc") |> dplyr::select(1,3,4,6)) |>
        tidyr::drop_na()
```

```
all_k79.filt.M |>
        dplyr::mutate(Sp = factor(Sp, levels = c(
                "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
        ))) |>
        dplyr::mutate(peak_K79 = case_when(
                peak_K79 == TRUE ~ "yes",
                .default = "no"
        )) |>
        dplyr::mutate(peak_K79 = factor(peak_K79, levels = c("yes", "no"))) |>
        ggplot2::ggplot(aes(x = peak_K79, colour = peak_K79, y = log2_TPM_plus_1)) |>
        plot_style() +
        ggplot2::facet_wrap(.~Sp, nrow = 1) +
        ggplot2::labs(x = "", y = "log2(TPM+1)", colour = "H3K79me2 mark") +
        ggplot2::theme(axis.text.x = element_blank(), axis.ticks.x = element_blank(),
                       legend.position = "top") +
        ggpubr::stat_compare_means(
                label = "p.adj",
                method = "wilcox.test",
                comparisons = list(c("yes", "no")),
                p.adjust.method = "bonferroni",
                tip.length = 0,
                size = 2.5
        ) +
        EnvStats::stat_n_text(size = 2.5)
```



For males

What this suggests is that we cannot look at the whole gene. From other data in the MS, we need to focus on the TSS.

TSS H3K79me2 marks

```
library(rtracklayer)
library(GenomicRanges)
library(ChIPpeakAnno)
library(reshape2)
```

```
# this is using
# /ebio/abt5_projects/PhaeoChromo/data/early-stage_evochromo/
# Ec
# Spro
# Und
# Dher
# Ddud
# Si
```

Due to issues in the gff files, I ran gffread on them.

```
# gffread --keep-genes -E Desmarestia-herbacea.reference.gt.AGAT.gff -o convert_gffread/Desmarestia-her
# gffread --keep-genes -E Scytosiphon-promiscuus_MALE.chr.gff -o convert_gffread/Scytosiphon-promiscuus
# gffread --keep-genes -E Schizocladia-ischiensis.AGAT.gtf -o convert_gffread/Schizocladia-ischiensis.A
```

We can load in the annotation.

```
genes.Ec <-
        rtracklayer::import.gff("data/annotation/Ec32_V5.longest_mRNA.SDR_F.AGAT.mod.gtf") |>
        subset(type == "mRNA")
genes.Und <-
        rtracklayer::import("data/annotation/PUBLIC_Undaria-pinnatifida_MALE.jv3.gtf") |>
        subset(type == "transcript")
genes.Ddud <-
        rtracklayer::import("data/annotation/Desmarestia-dudresnayi.cleaned2025.jv.gffread.gtf")  |>
        subset(type == "transcript")
genes.Spro <-
        rtracklayer::import("data/annotation/convert_gffread/Scytosiphon-promiscuus_MALE.chr.gffread.gf
        subset(type == "mRNA")
genes.Dher <-
        rtracklayer::import("data/annotation/convert_gffread/Desmarestia-herbacea.reference.gt.AGAT.gff
        subset(type == "mRNA")
genes.Si <-
        rtracklayer::import("data/annotation/convert_gffread/Schizocladia-ischiensis.AGAT.gffread.gtf",
        subset(type == "gene")
```

Now, load in the MACS2 peaks

```
###### ----------- load in the MACS2 peaks
path <- "data/K79_data/"
files <- dir(path, "Peak")
files.Ddud <- files[1]
files.Dher.f <- files[2]
files.Dher.m <- files[3]
files.Ec.f <- files[4]
files.Ec.m <- files[5]
files.Si <- files[6]
files.Spro.f <- files[7]
files.Spro.m <- files[8]
files.Und.f <- files[9]
files.Und.m <- files[10]
```

```
peak_list.Ddud <- sapply(file.path(path, files.Ddud), toGRanges, format="broadPeak")
peak_list.Dher.f <- sapply(file.path(path, files.Dher.f), toGRanges, format="broadPeak")
peak_list.Dher.m <- sapply(file.path(path, files.Dher.m), toGRanges, format="broadPeak")
peak_list.Ec.f <- sapply(file.path(path, files.Ec.f), toGRanges, format="broadPeak")
peak_list.Ec.m <- sapply(file.path(path, files.Ec.m), toGRanges, format="broadPeak")
```

```r
peak_list.Si <- sapply(file.path(path, files.Si), toGRanges, format="broadPeak")
peak_list.Spro.f <- sapply(file.path(path, files.Spro.f), toGRanges, format="broadPeak")
peak_list.Spro.m <- sapply(file.path(path, files.Spro.m), toGRanges, format="broadPeak")
peak_list.Und.f <- sapply(file.path(path, files.Und.f), toGRanges, format="broadPeak")
peak_list.Und.m <- sapply(file.path(path, files.Und.m), toGRanges, format="broadPeak")
```

Function to find the overlaps on the TSS

```r
library(GenomicRanges)
# genes_Sp = genes.Und
# peak_list_Sp = peak_list.Und

find_overlap_tss <- function(genes_Sp, peak_list_Sp, window_around_tss = 1){
        # 1. Get TSS from gene annotations
        TSS <- resize(genes_Sp, width = 1, fix = "start")  # TSS is start of gene regardless of strand
        TSS[strand(TSS) == "-"] <- resize(TSS[strand(TSS) == "-"], width = 1, fix = "end")  # adjust fo

        # 2. Define a window around TSS (e.g. ±500 bp)
        TSS_window <- resize(TSS, width = window_around_tss, fix = "center")  # +/-500 bp

        # 3. Extract the GRanges of peaks
        peaks <- peak_list_Sp[[1]]

        # 4. Find overlaps
        hits <- findOverlaps(peaks, TSS_window)

        # 5. Get overlapping peaks and/or genes
        # overlapping_peaks <- peaks[queryHits(hits)]
        overlapping_genes <- genes_Sp[subjectHits(hits)]
        return(overlapping_genes)
}
```

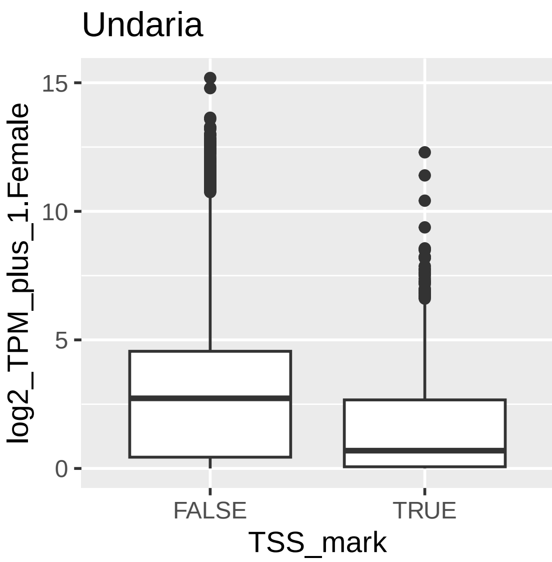
Find overlaps on the TSS.

```r
overlap_Ec.f <- find_overlap_tss(genes.Ec, peak_list.Ec.f, window_around_tss = 1)
overlap_Ec.m <- find_overlap_tss(genes.Ec, peak_list.Ec.m, window_around_tss = 1)
overlap_Spro.f <- find_overlap_tss(genes.Spro, peak_list.Spro.f, window_around_tss = 1)
overlap_Spro.m <- find_overlap_tss(genes.Spro, peak_list.Spro.m, window_around_tss = 1)
overlap_Und.f <- find_overlap_tss(genes.Und, peak_list.Und.f, window_around_tss = 1)
overlap_Und.m <- find_overlap_tss(genes.Und, peak_list.Und.m, window_around_tss = 1)
overlap_Dher.f <- find_overlap_tss(genes.Dher, peak_list.Dher.f, window_around_tss = 1)
overlap_Dher.m <- find_overlap_tss(genes.Dher, peak_list.Dher.m, window_around_tss = 1)
overlap_Ddud <- find_overlap_tss(genes.Ddud, peak_list.Ddud, window_around_tss = 1)
overlap_Si <- find_overlap_tss(genes.Si, peak_list.Si, window_around_tss = 1)


gene_TSS.Ec.f <- overlap_Ec.f$ID
gene_TSS.Ec.m <- overlap_Ec.m$ID
gene_TSS.Spro.f <- overlap_Spro.f$ID
gene_TSS.Spro.m <- overlap_Spro.m$ID
gene_TSS.Und.f <- overlap_Und.f$gene_id
gene_TSS.Und.m <- overlap_Und.m$gene_id
gene_TSS.Dher.f <- overlap_Dher.f$ID
gene_TSS.Dher.m <- overlap_Dher.m$ID
gene_TSS.Ddud <- overlap_Ddud$gene_id
```

```r
gene_TSS.Si <- overlap_Si$ID
```

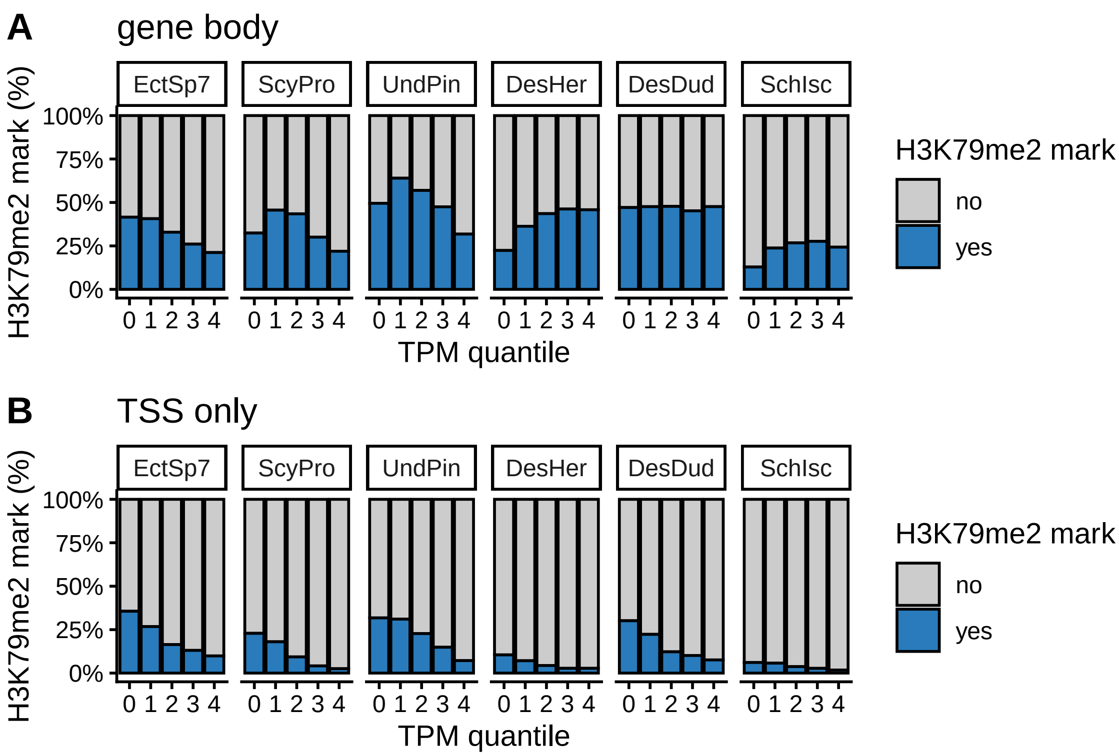
```r
Ec_k79.tss.f <-
        Ec_k79 |>
        dplyr::mutate(TSS_mark = case_when(gene_id %in% gene_TSS.Ec.f ~ TRUE, .default = FALSE))
Ec_k79.tss.m <-
        Ec_k79 |>
        dplyr::mutate(TSS_mark = case_when(gene_id %in% gene_TSS.Ec.m ~ TRUE, .default = FALSE))
Spro_k79.tss.f <-
        Spro_k79 |>
        dplyr::mutate(TSS_mark = case_when(gene_id %in% gene_TSS.Spro.f ~ TRUE, .default = FALSE))
Spro_k79.tss.m <-
        Spro_k79 |>
        dplyr::mutate(TSS_mark = case_when(gene_id %in% gene_TSS.Spro.m ~ TRUE, .default = FALSE))
Und_k79.tss.f <-
        Und_k79 |>
        dplyr::mutate(TSS_mark = case_when(gene_id %in% gene_TSS.Und.f ~ TRUE, .default = FALSE))
Und_k79.tss.m <-
        Und_k79 |>
        dplyr::mutate(TSS_mark = case_when(gene_id %in% gene_TSS.Und.m ~ TRUE, .default = FALSE))
Dher_k79.tss.f <-
        Dher_k79 |>
        dplyr::mutate(TSS_mark = case_when(gene_id %in% gene_TSS.Dher.f ~ TRUE, .default = FALSE))
Dher_k79.tss.m <-
        Dher_k79 |>
        dplyr::mutate(TSS_mark = case_when(gene_id %in% gene_TSS.Dher.m ~ TRUE, .default = FALSE))
Ddud_k79.tss <-
        Ddud_k79 |>
        dplyr::mutate(TSS_mark = case_when(gene_id %in% gene_TSS.Ddud ~ TRUE, .default = FALSE))
Si_k79.tss <-
        Si_k79 |>
        dplyr::mutate(TSS_mark = case_when(gene_id %in% gene_TSS.Si ~ TRUE, .default = FALSE))

all_k79.filt.TPM.tss <-
        Ec_k79.tss.f |> dplyr::mutate(Sp = "EctSp7") |>
        rbind(Spro_k79.tss.f |> dplyr::mutate(Sp = "ScyPro")) |>
        rbind(Und_k79.tss.f |> dplyr::mutate(Sp = "UndPin")) |>
        rbind(Dher_k79.tss.f |> dplyr::mutate(Sp = "DesHer")) |>
        dplyr::select(gene_id,
                      TPM = TPM.Female,
                      peak_K79 = peak_K79.Female,
                      Sp, TSS_mark) |>
        rbind(Ddud_k79.tss |> dplyr::mutate(Sp = "DesDud") |> dplyr::select(1,2,4,6,7)) |>
        rbind(Si_k79.tss |> dplyr::mutate(Sp = "SchIsc") |> dplyr::select(1,2,4,6,7)) |>
        tidyr::drop_na()
```

```r
# practice
Und_k79 |>
        dplyr::mutate(TSS_mark = case_when(gene_id %in% gene_TSS.Und.f ~ TRUE, .default = FALSE)) |>
        ggplot2::ggplot(aes(x = TSS_mark, y = log2_TPM_plus_1.Female)) +
        ggplot2::geom_boxplot() +
        ggplot2::labs(title = "Undaria")
```




```
# full data
TPM_threshold = 0
all_k79.filt.TPM.filt.tss <-
        all_k79.filt.TPM.tss |>
        dplyr::group_by(Sp) |>
        dplyr::mutate(
                quant_tmp = {
                        x = TPM
                        x[TPM > TPM_threshold] = ntile(x[TPM > TPM_threshold], 4)
                        x[TPM <= TPM_threshold] = NA_integer_
                        x
                },
                TPM_quantile = dplyr::case_when(
                        TPM <= TPM_threshold ~ "0",
                        TRUE ~ as.character(quant_tmp)
                )
        ) |>
        dplyr::select(-quant_tmp) |>
        dplyr::ungroup() |>
        dplyr::mutate(Sp = factor(Sp, levels = c(
                "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
        )))
# dplyr::mutate(TPM_quantile = dplyr::case_when(
#         TPM == 0 ~ 0,
#         TPM > 0 ~ ntile(TPM, 4)
# ),
# .by = Sp)

p_1 <- all_k79.filt.TPM.filt.tss |>
        dplyr::mutate(peak_K79 = case_when(
                peak_K79 == TRUE ~ "yes",
                .default = "no"
        )) |>
        # dplyr::mutate(peak_K79 = factor(peak_K79, levels = c("yes", "no"))) |>
        ggplot2::ggplot(aes(x = factor(TPM_quantile), fill = peak_K79)) +
        ggplot2::geom_bar(position = "fill", colour = "black") +
```

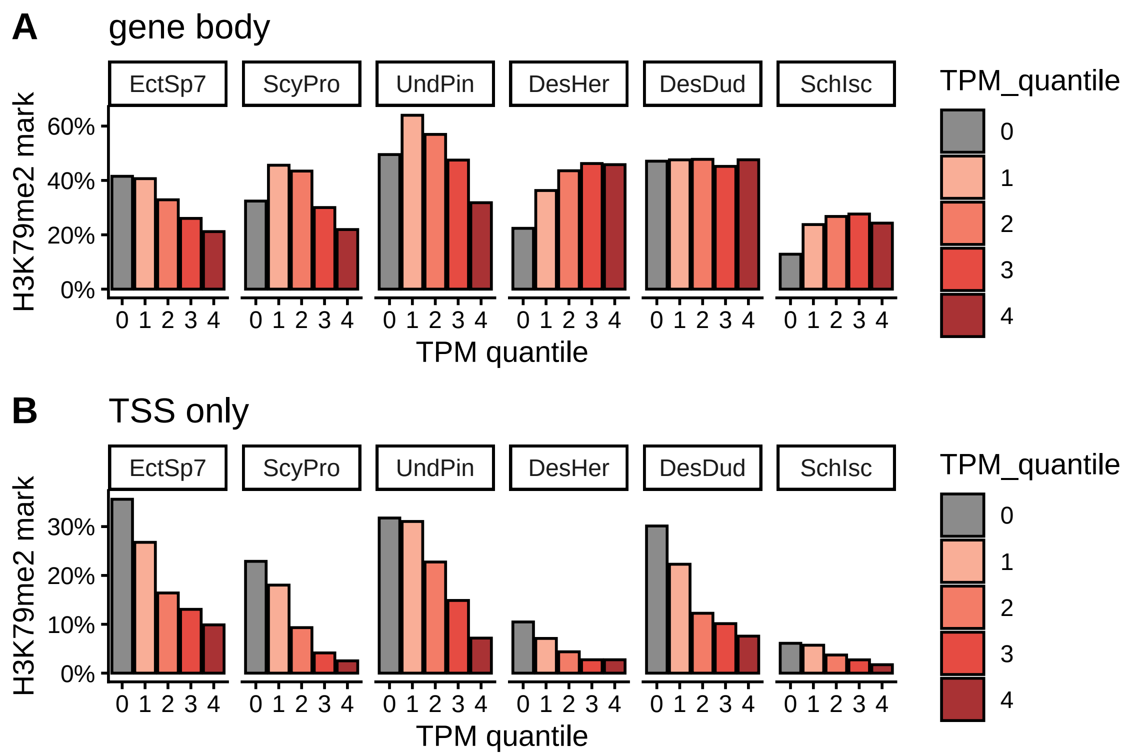
```r
        ggplot2::scale_y_continuous(labels = scales::percent_format()) +
        ggplot2::labs(
                x = "TPM quantile",
                y = "H3K79me2 mark (%)",
                fill = "H3K79me2 mark",
                title = "gene body"
        ) +
        ggplot2::theme_classic() +
        ggplot2::scale_fill_manual(values = c("grey80", "#297BBB")) +
        ggplot2::facet_wrap(.~Sp, nrow = 1)

p_2 <- all_k79.filt.TPM.filt.tss |>
        dplyr::mutate(TSS_mark = case_when(
                TSS_mark == TRUE ~ "yes",
                .default = "no"
        )) |>
        ggplot2::ggplot(aes(x = factor(TPM_quantile), fill = TSS_mark)) +
        ggplot2::geom_bar(position = "fill", colour = "black") +
        ggplot2::scale_y_continuous(labels = scales::percent_format()) +
        ggplot2::labs(
                x = "TPM quantile",
                y = "H3K79me2 mark (%)",
                fill = "H3K79me2 mark",
                title = "TSS only"
        ) +
        ggplot2::theme_classic() +
        ggplot2::scale_fill_manual(values = c("grey80", "#297BBB")) +
        ggplot2::facet_wrap(.~Sp, nrow = 1)

ggpubr::ggarrange(p_1,p_2,nrow = 2, labels = c("A","B"))
```




```r
# ggplot2::ggsave("figure/K79_tss_vs_gene_body_TPM_quantiles_barplot2.pdf", width = 6, height = 4)
```

```r
p_1 <- all_k79.filt.TPM.filt.tss |>
        dplyr::mutate(peak_K79 = case_when(
                peak_K79 == TRUE ~ "yes",
                .default = "no"
        )) |>
        dplyr::mutate(n_genes = dplyr::n(), .by = c(Sp, TPM_quantile)) |>
        dplyr::filter(peak_K79 == "yes") |>
        # compute percentage
        dplyr::summarise(n_mark = dplyr::n(), .by = c(Sp, TPM_quantile, n_genes)) |>
        dplyr::mutate(p_mark = n_mark/n_genes) |>
        ggplot2::ggplot(aes(x = TPM_quantile, y = p_mark, fill = TPM_quantile)) +
        ggplot2::geom_col(colour = "black") +
        ggplot2::scale_y_continuous(labels = scales::percent_format()) +
        ggplot2::scale_fill_manual(values = c("#8B8B8B", "#F9AE97", "#F37C67", "#E64B42", "#A93234")) +
        ggplot2::labs(
                x = "TPM quantile",
                y = "H3K79me2 mark",
                title = "gene body"
        ) +
        ggplot2::theme_classic() +
        ggplot2::facet_wrap(.~Sp, nrow = 1)

p_2 <- all_k79.filt.TPM.filt.tss |>
        dplyr::mutate(TSS_mark = case_when(
                TSS_mark == TRUE ~ "yes",
                .default = "no"
        )) |>
        dplyr::mutate(n_genes = dplyr::n(), .by = c(Sp, TPM_quantile)) |>
```

```
        dplyr::filter(TSS_mark == "yes") |>
        # compute percentage
        dplyr::summarise(n_mark = dplyr::n(), .by = c(Sp, TPM_quantile, n_genes)) |>
        dplyr::mutate(p_mark = n_mark/n_genes) |>
        ggplot2::ggplot(aes(x = TPM_quantile, y = p_mark, fill = TPM_quantile)) +
        ggplot2::geom_col(colour = "black") +
        ggplot2::scale_y_continuous(labels = scales::percent_format()) +
        ggplot2::scale_fill_manual(values = c("#8B8B8B", "#F9AE97", "#F37C67", "#E64B42", "#A93234")) +
        ggplot2::labs(
                x = "TPM quantile",
                y = "H3K79me2 mark",
                title = "TSS only"
        ) +
        ggplot2::theme_classic() +
        ggplot2::facet_wrap(.~Sp, nrow = 1)

ggpubr::ggarrange(p_1,p_2,nrow = 2, labels = c("A","B"))
```




```
# ggplot2::ggsave("figure/K79_tss_vs_gene_body_TPM_quantiles_barplot3.pdf", width = 6, height = 4)
```

```
# TSS mark
stats_0_vs_rest <- all_k79.filt.TPM.filt.tss |>
        dplyr::mutate(peak_K79 = if_else(peak_K79, TRUE, FALSE)) |>
        dplyr::mutate(group = if_else(TPM_quantile == "0", "Q0", "Rest")) |>
        dplyr::summarise(
                n_mark  = sum(peak_K79),
                n_total = n(),
                .by = c(Sp, group)
        )
```

Sp	p_value	p_adj
EctSp7	8.893×10^{-3}	5.336×10^{-2}
ScyPro	7.902×10^{-2}	4.741×10^{-1}
UndPin	6.277×10^{-1}	1.000
DesHer	4.544×10^{-42}	2.726×10^{-41}
DesDud	9.670×10^{-1}	1.000
SchIsc	6.438×10^{-15}	3.863×10^{-14}

```
stats_0_vs_rest_pvals <- stats_0_vs_rest |>
        tidyr::pivot_wider(names_from = group, values_from = c(n_mark, n_total)) |>
        dplyr::mutate(
                p_value = purrr::pmap_dbl(
                        list(n_mark_Q0, n_total_Q0, n_mark_Rest, n_total_Rest),
                        function(a, b, c, d) {
                                matrix_data <- matrix(c(a, b - a, c, d - c), nrow = 2)
                                fisher.test(matrix_data)$p.value
                        }
                ),
                p_adj = p.adjust(p_value, method = "bonferroni")
        ) |>
        dplyr::select(Sp, p_value, p_adj)
stats_0_vs_rest_pvals |>
        gt::gt() |>
        gt::fmt_scientific(
                columns = c(p_value, p_adj),
                decimals = 3
        )
```

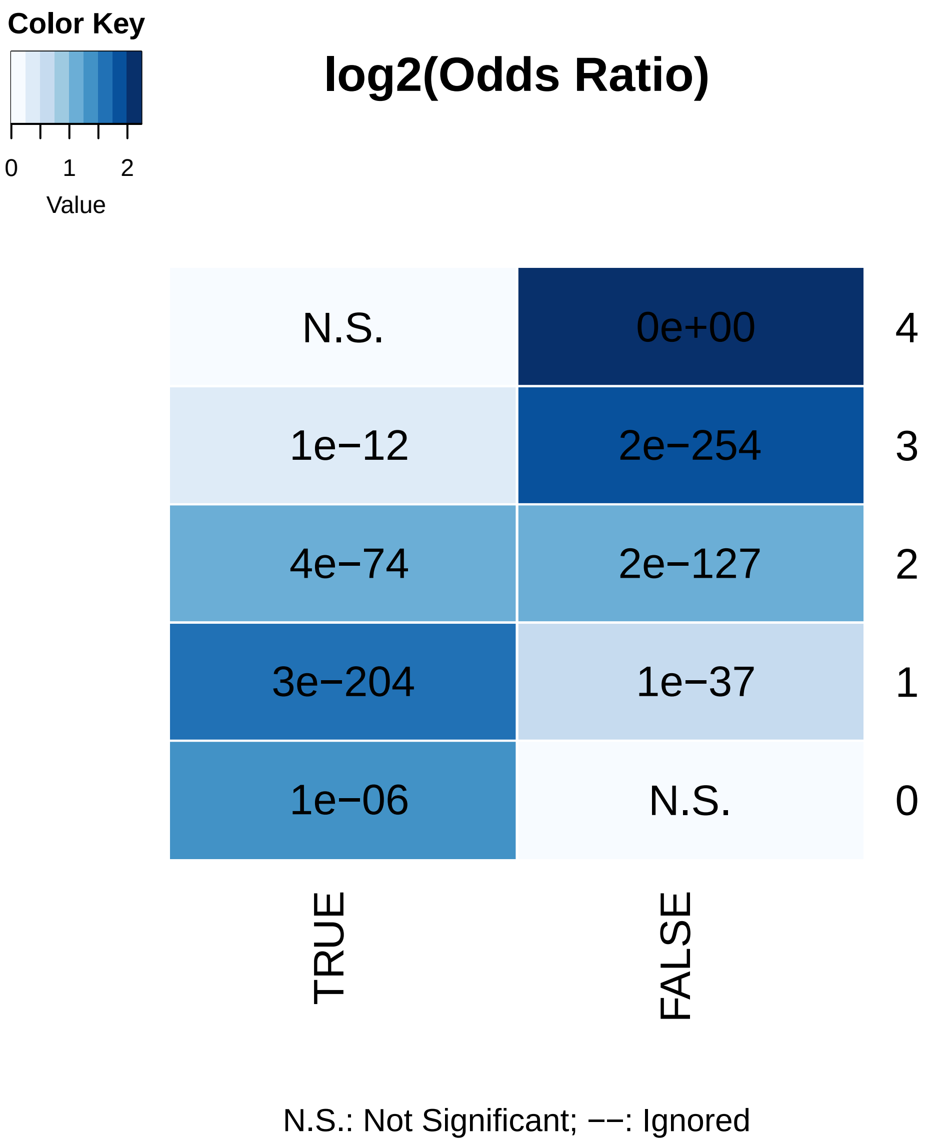
```
# TSS mark
stats_0_vs_rest <- all_k79.filt.TPM.filt.tss |>
        dplyr::mutate(TSS_mark = if_else(TSS_mark, TRUE, FALSE)) |>
        dplyr::mutate(group = if_else(TPM_quantile == "0", "Q0", "Rest")) |>
        dplyr::summarise(
                n_mark  = sum(TSS_mark),
                n_total = n(),
                .by = c(Sp, group)
        )

stats_0_vs_rest_pvals <- stats_0_vs_rest |>
        tidyr::pivot_wider(names_from = group, values_from = c(n_mark, n_total)) |>
        dplyr::mutate(
                p_value = purrr::pmap_dbl(
                        list(n_mark_Q0, n_total_Q0, n_mark_Rest, n_total_Rest),
                        function(a, b, c, d) {
                                matrix_data <- matrix(c(a, b - a, c, d - c), nrow = 2)
                                fisher.test(matrix_data)$p.value
                        }
                ),
                p_adj = p.adjust(p_value, method = "bonferroni")
        ) |>
        dplyr::select(Sp, p_value, p_adj)
```

Sp	p_value	p_adj
EctSp7	6.158×10^{-7}	3.695×10^{-6}
ScyPro	6.648×10^{-38}	3.989×10^{-37}
UndPin	1.276×10^{-42}	7.657×10^{-42}
DesHer	3.267×10^{-16}	1.960×10^{-15}
DesDud	6.097×10^{-99}	3.658×10^{-98}
SchIsc	1.021×10^{-3}	6.125×10^{-3}

```
stats_0_vs_rest_pvals |>
        gt::gt() |>
        gt::fmt_scientific(
                columns = c(p_value, p_adj),
                decimals = 3
        )
```

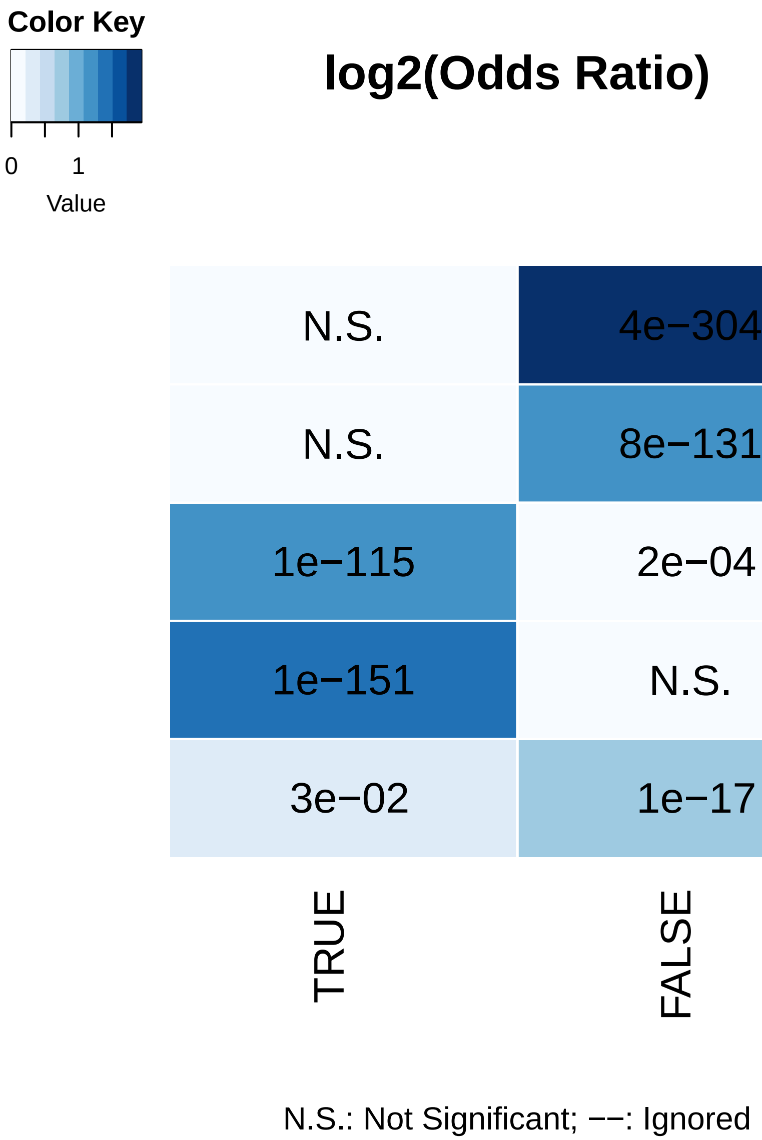
Plot this data as a heatmap with percentage of genes with mark in each TPM quantile.

```
# using the GeneOverlap package to test for significance of overlap between genes with TSS mark and gen
library(GeneOverlap)
all_k79.filt.TPM.filt.tss_TPM_quantile_list <-
        all_k79.filt.TPM.filt.tss |>
        summarise(gene_id = list(gene_id), .by = c(TPM_quantile, Sp)) |>
        dplyr::mutate(TPM_quantile = factor(TPM_quantile, levels = rev(c("0","1","2","3","4")))) |>
        dplyr::arrange(TPM_quantile)

all_k79.filt.TPM.filt.tss_peak79_list <-
        all_k79.filt.TPM.filt.tss |>
        summarise(gene_id = list(gene_id), .by = c(peak_K79, Sp)) |>
        dplyr::mutate(peak_K79 = factor(peak_K79, levels = c(TRUE, FALSE))) |>
        dplyr::arrange(peak_K79)

# gene body EctSp7
Ec_body <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "EctSp7") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_peak79_list |>
                dplyr::filter(Sp == "EctSp7") |>
                dplyr::select(-Sp) |>
                tibble::deframe())

Ec_body |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



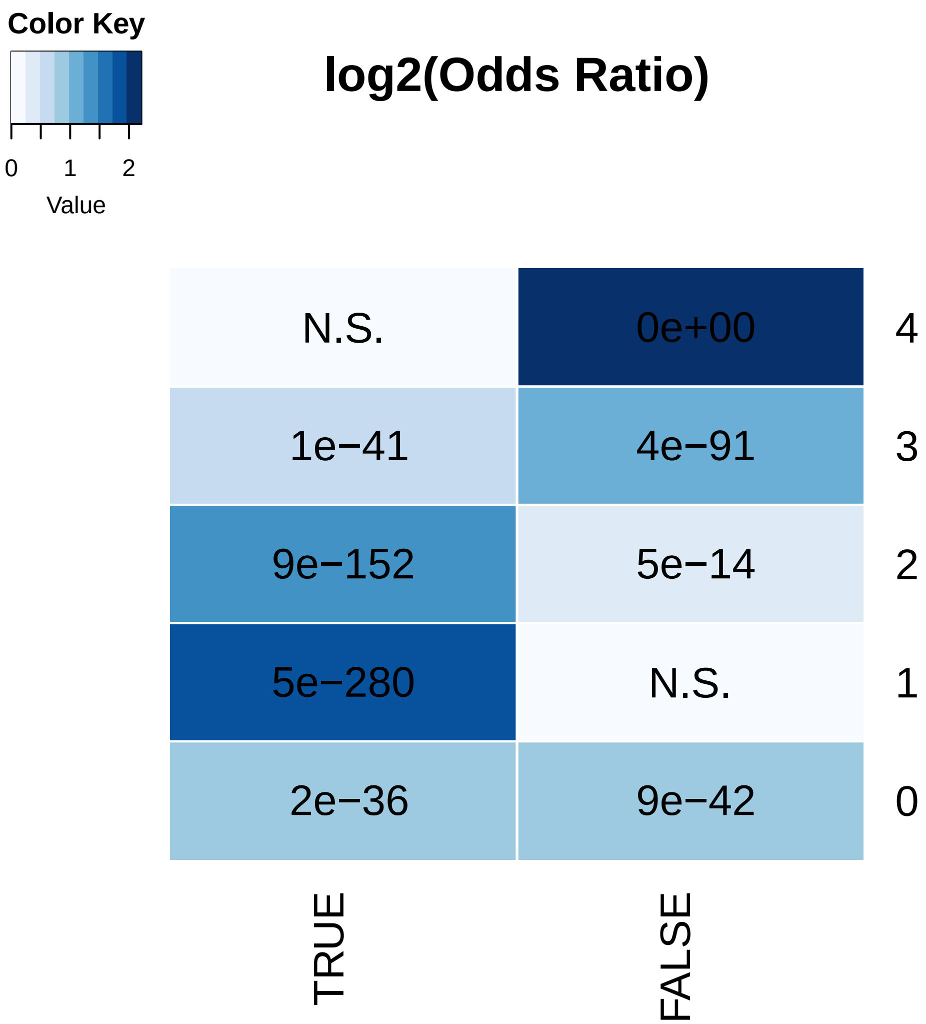

```
# Ec_body |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ec_body_pval.tsv")
# Ec_body |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ec_body_odds.ratio.tsv")

# gene body Spro
Spro_body <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "ScyPro") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
```

```r
        all_k79.filt.TPM.filt.tss_peak79_list |>
                dplyr::filter(Sp == "ScyPro") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Spro_body |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```




```r
# Spro_body |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Spro_body_pval.tsv")
# Spro_body |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
```

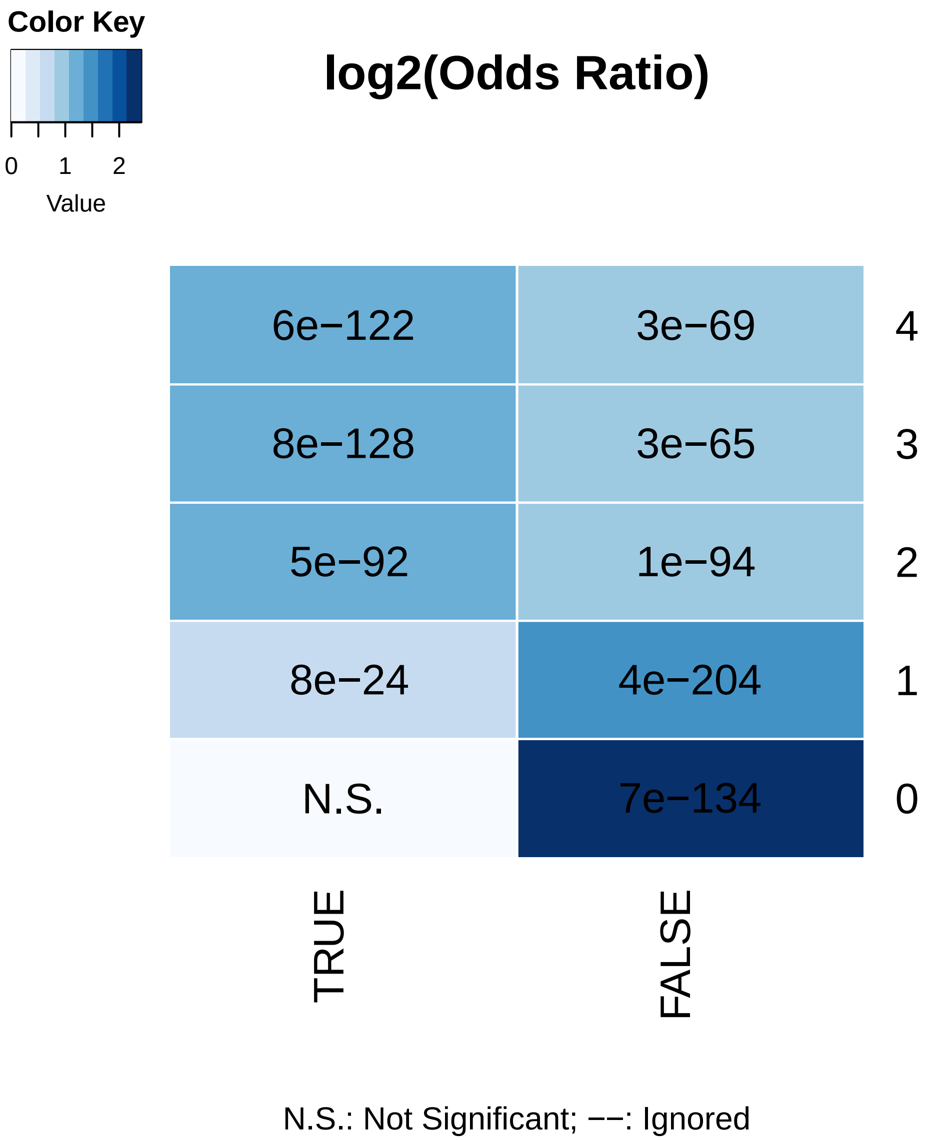
```r
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Spro_body_odds.ratio.tsv")

# gene body UndPin
Upin_body <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "UndPin") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_peak79_list |>
                dplyr::filter(Sp == "UndPin") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Upin_body |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



N.S.: Not Significant; −−: Ignored

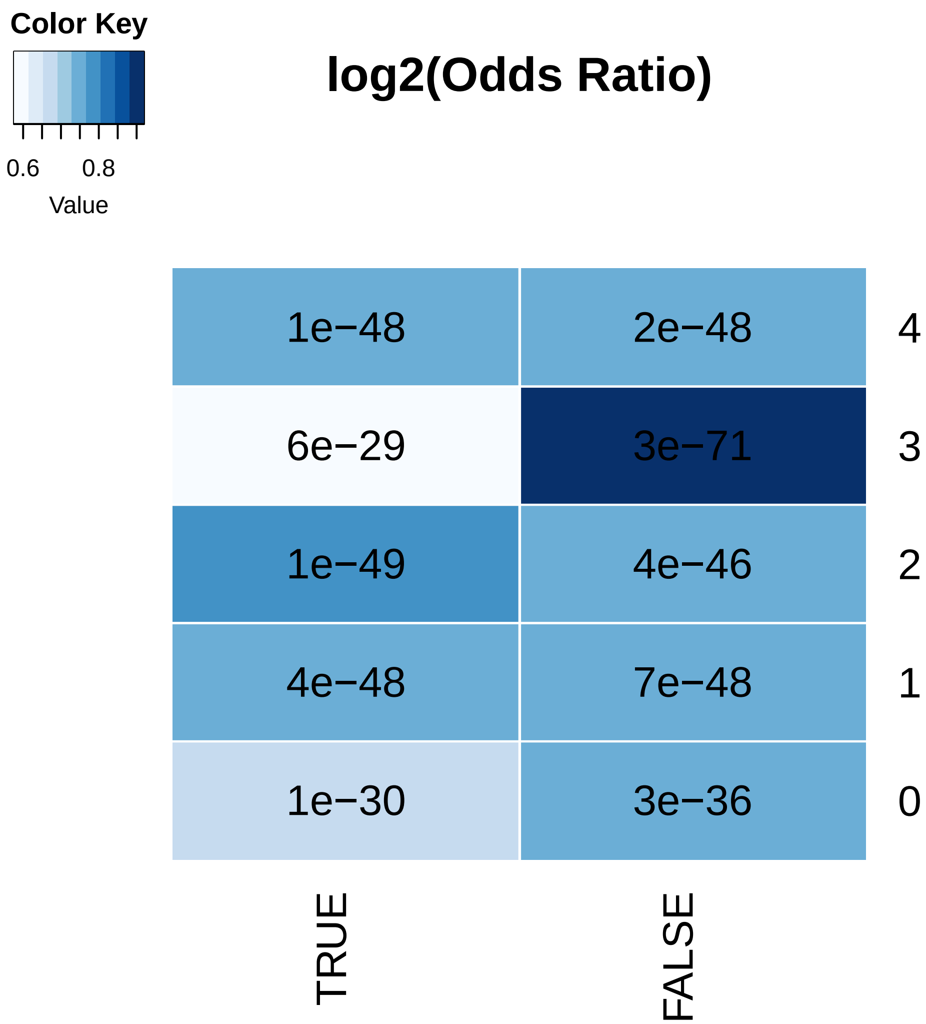
```
# Upin_body |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Upin_body_pval.tsv")
# Upin_body |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Upin_body_odds.ratio.tsv")

# gene body DesHer
Dher_body <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "DesHer") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
```

```
        all_k79.filt.TPM.filt.tss_peak79_list |>
                dplyr::filter(Sp == "DesHer") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Dher_body |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```




```
# Dher_body |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Dher_body_pval.tsv")
# Dher_body |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
```

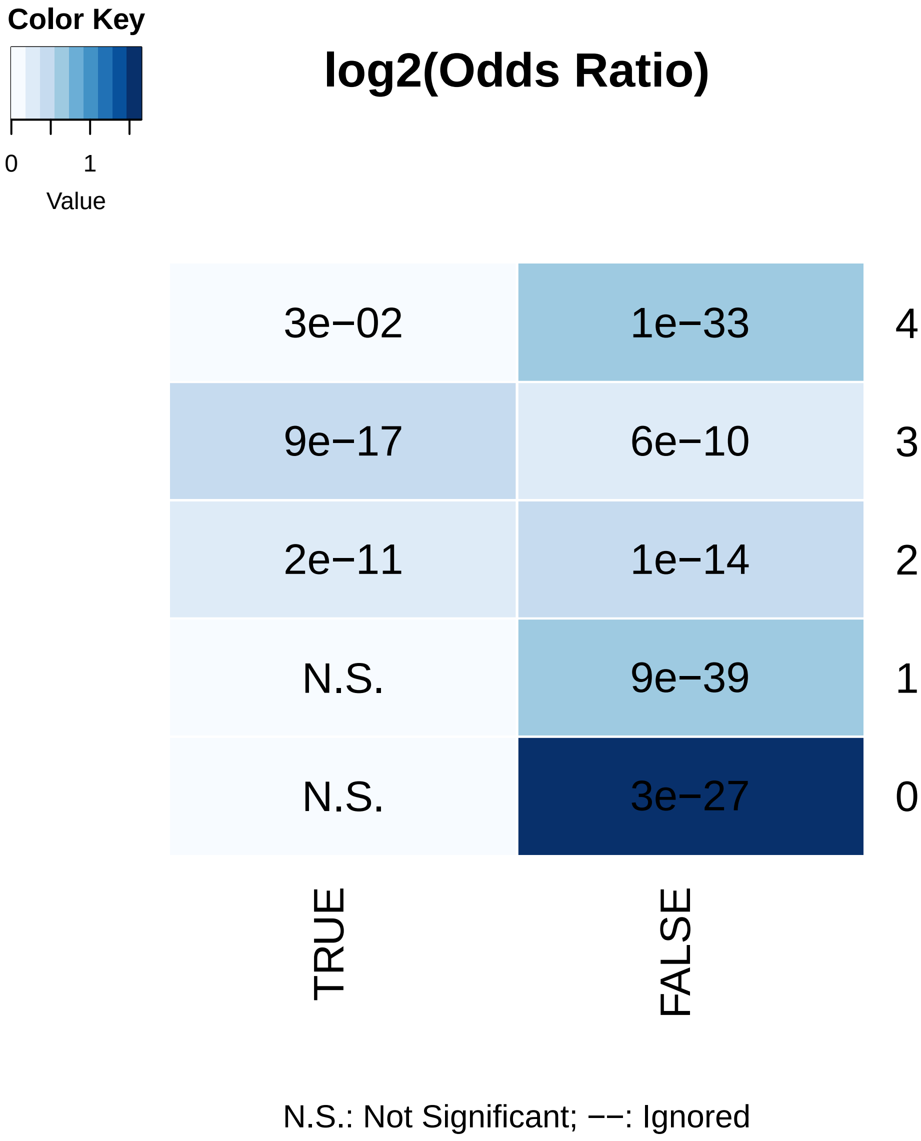
```
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Dher_body_odds.ratio.tsv")
# gene body DesDud
Ddud_body <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "DesDud") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_peak79_list |>
                dplyr::filter(Sp == "DesDud") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Ddud_body |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



N.S.: Not Significant; −−: Ignored

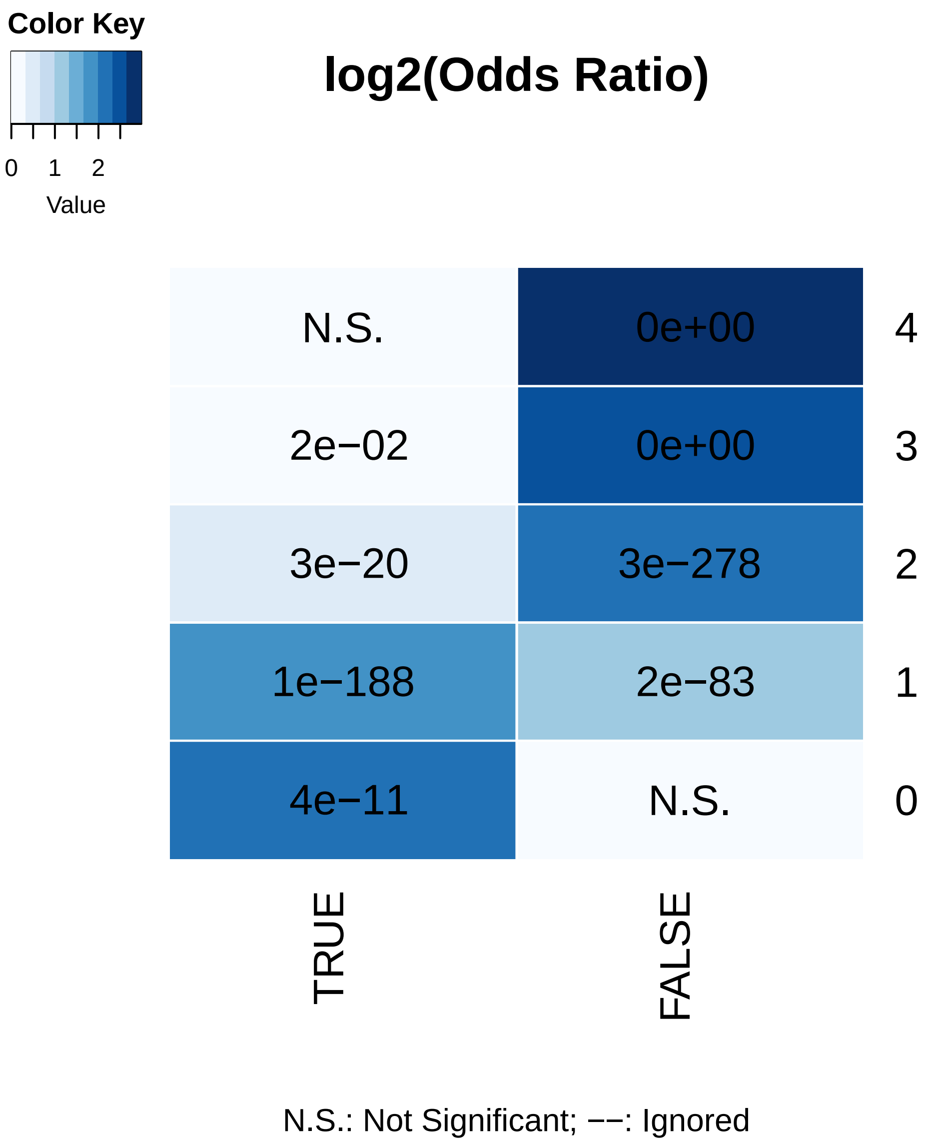
```r
# Ddud_body |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ddud_body_pval.tsv")
# Ddud_body |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ddud_body_odds.ratio.tsv")
# gene body SchIsc
Sisc_body <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "SchIsc") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_peak79_list |>
```

```
                 dplyr::filter(Sp == "SchIsc") |>
                 dplyr::select(-Sp) |>
                 tibble::deframe())
Sisc_body |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



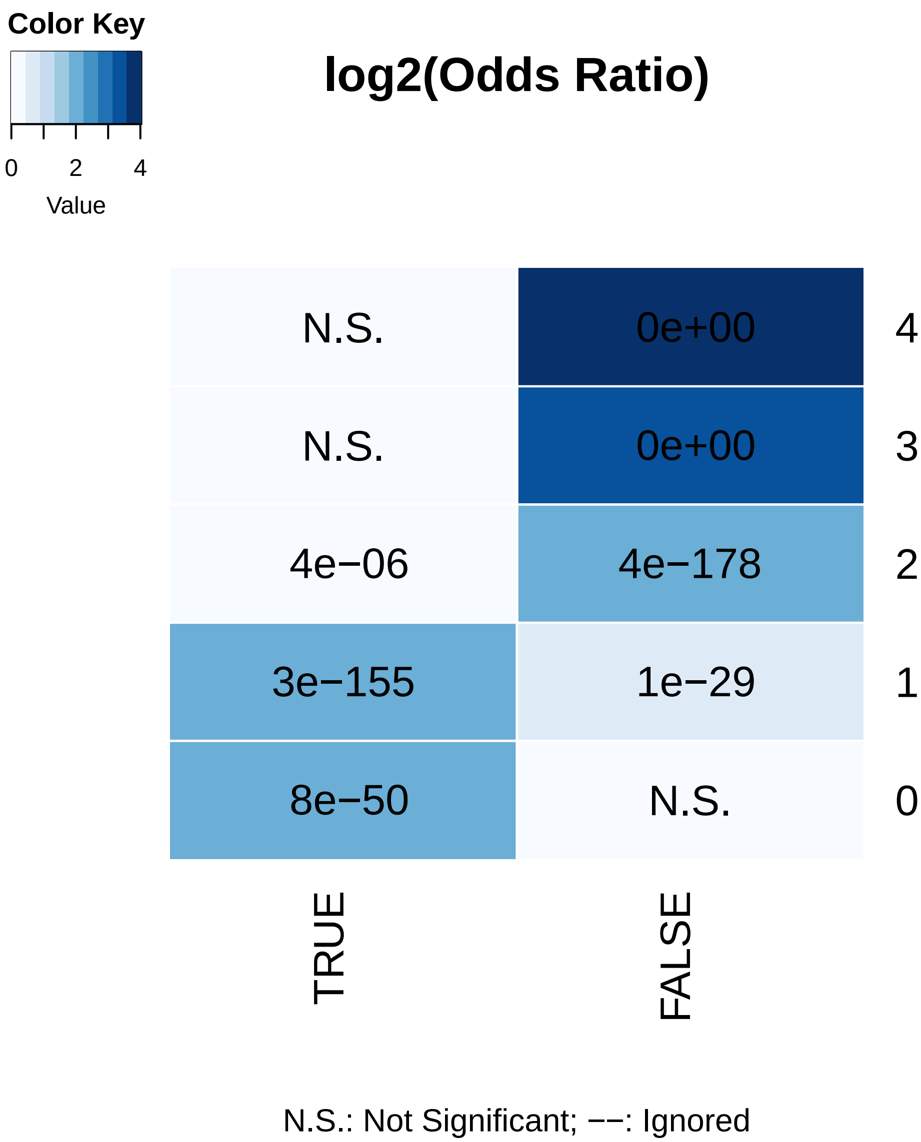

```
# Sisc_body |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Sisc_body_pval.tsv")
# Sisc_body |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
```

```r
#          tibble::rownames_to_column("TPM_quantile") |>
#          readr::write_tsv("table/Sisc_body_odds.ratio.tsv")

all_k79.filt.TPM.filt.tss_TSS_mark_list <-
        all_k79.filt.TPM.filt.tss |>
        summarise(gene_id = list(gene_id), .by = c(TSS_mark, Sp)) |>
        dplyr::mutate(TSS_mark = factor(TSS_mark, levels = c(TRUE, FALSE))) |>
        dplyr::arrange(TSS_mark)

# TSS EctSp7
Ec_tss <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "EctSp7") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_TSS_mark_list |>
                dplyr::filter(Sp == "EctSp7") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Ec_tss |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



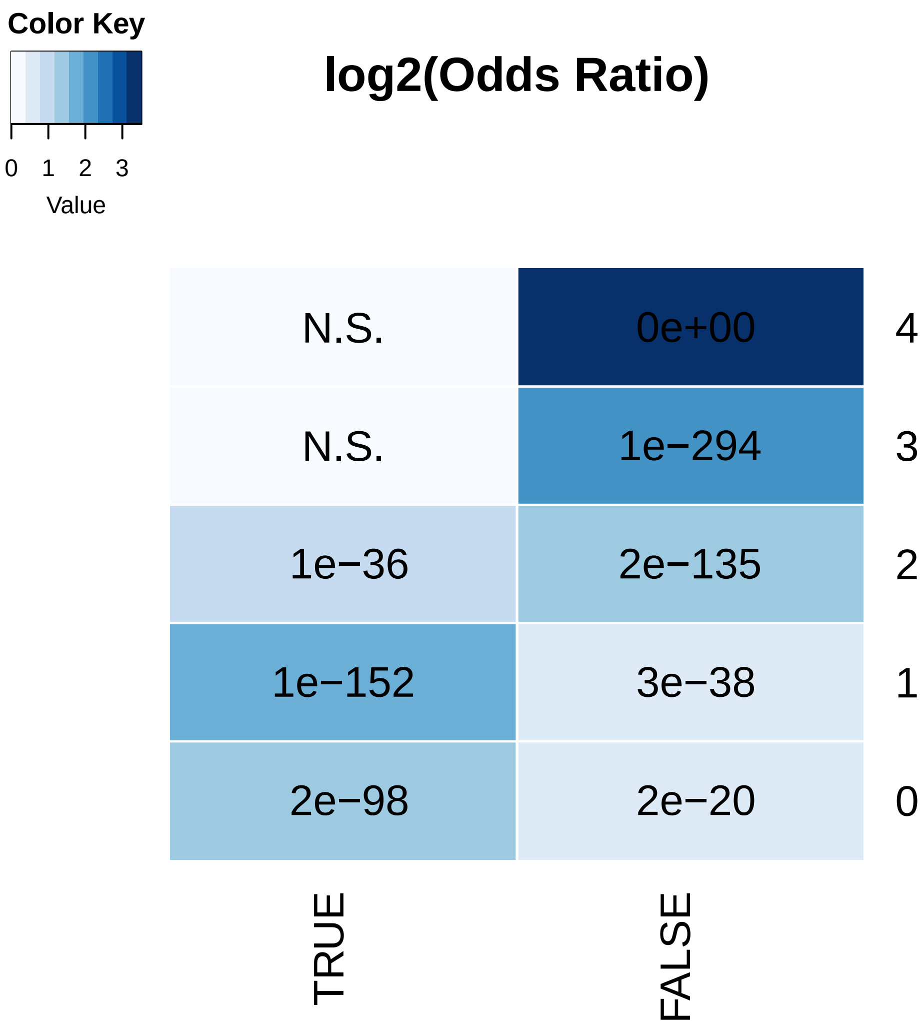

```r
# Ec_tss |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ec_tss_pval.tsv")
# Ec_tss |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ec_tss_odds.ratio.tsv")
# TSS Spro
Spro_tss <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "ScyPro") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_TSS_mark_list |>
```

```
                 dplyr::filter(Sp == "ScyPro") |>
                 dplyr::select(-Sp) |>
                 tibble::deframe())
Spro_tss |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```




```
# Spro_tss |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Spro_tss_pval.tsv")
# Spro_tss |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
```

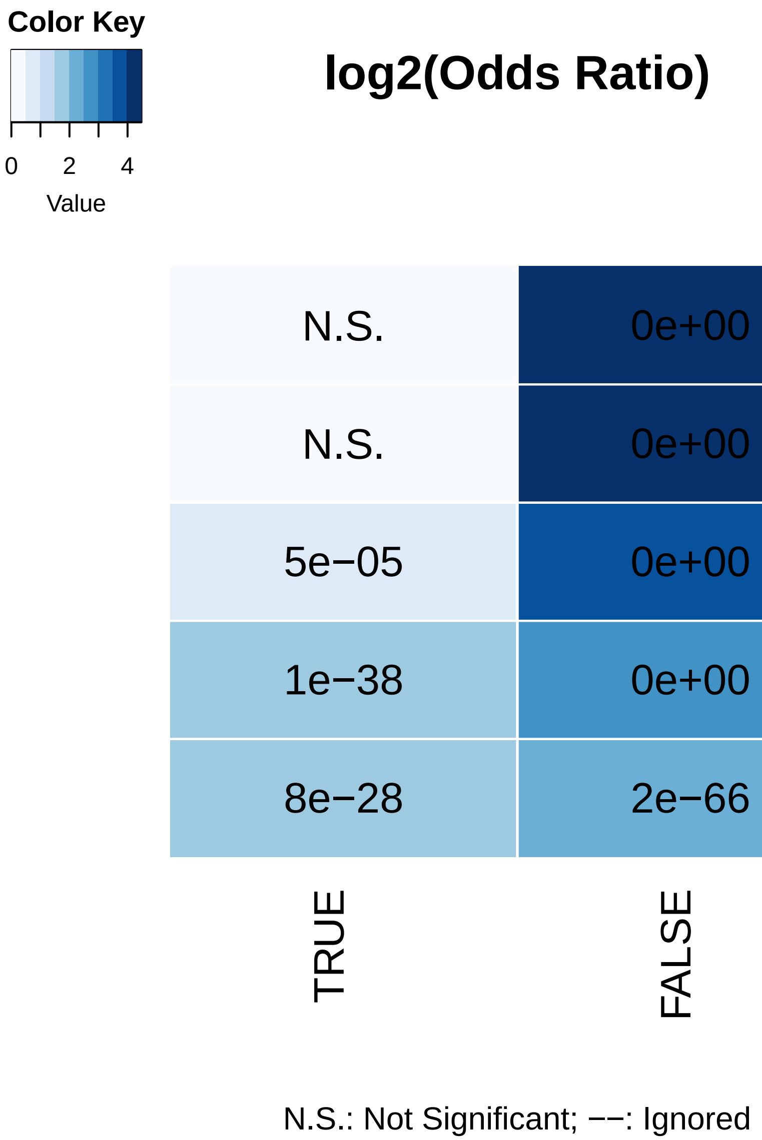
```r
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Spro_tss_odds.ratio.tsv")
# TSS UndPin
Upin_tss <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "UndPin") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_TSS_mark_list |>
                dplyr::filter(Sp == "UndPin") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Upin_tss |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



N.S.: Not Significant; −−: Ignored

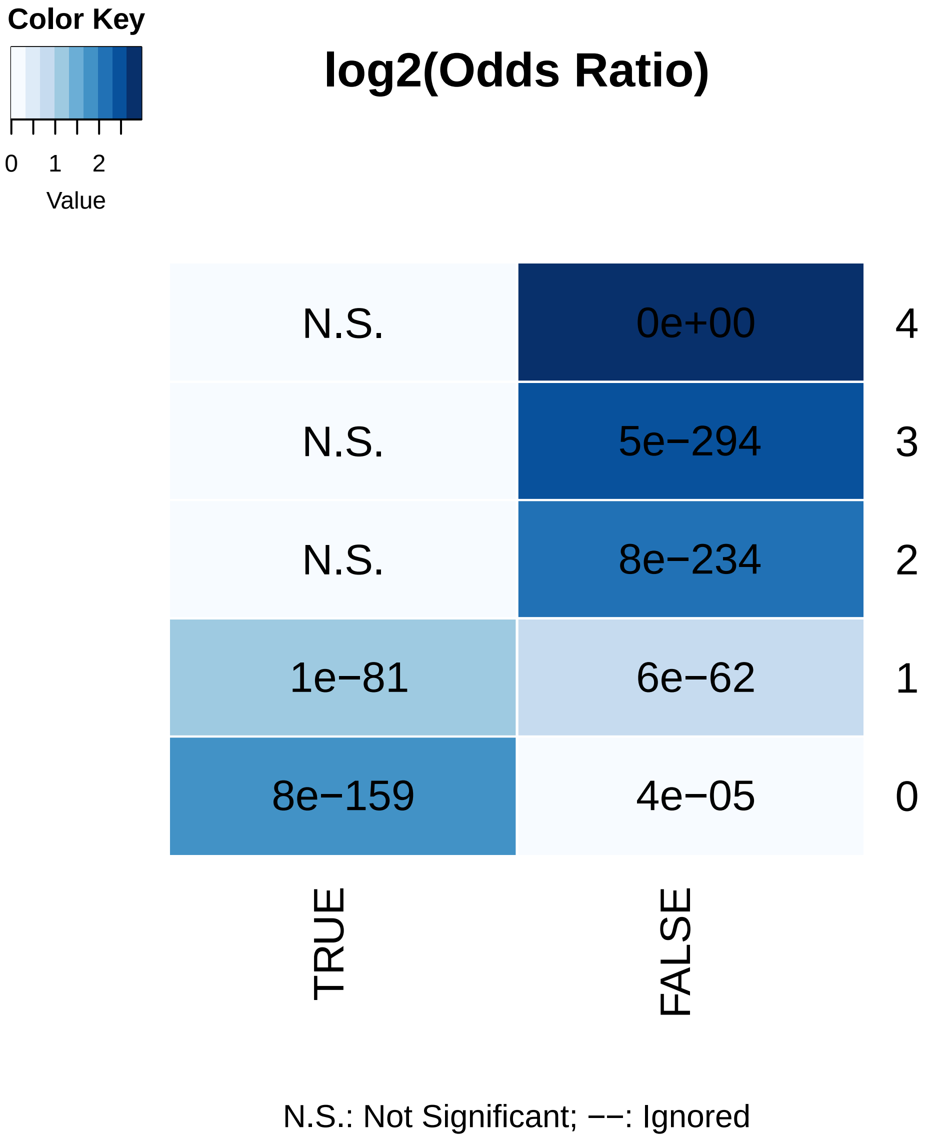
```
# Upin_tss |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Upin_tss_pval.tsv")
# Upin_tss |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Upin_tss_odds.ratio.tsv")

# TSS DesHer
Dher_tss <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "DesHer") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
```

```
        all_k79.filt.TPM.filt.tss_TSS_mark_list |>
                dplyr::filter(Sp == "DesHer") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Dher_tss |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



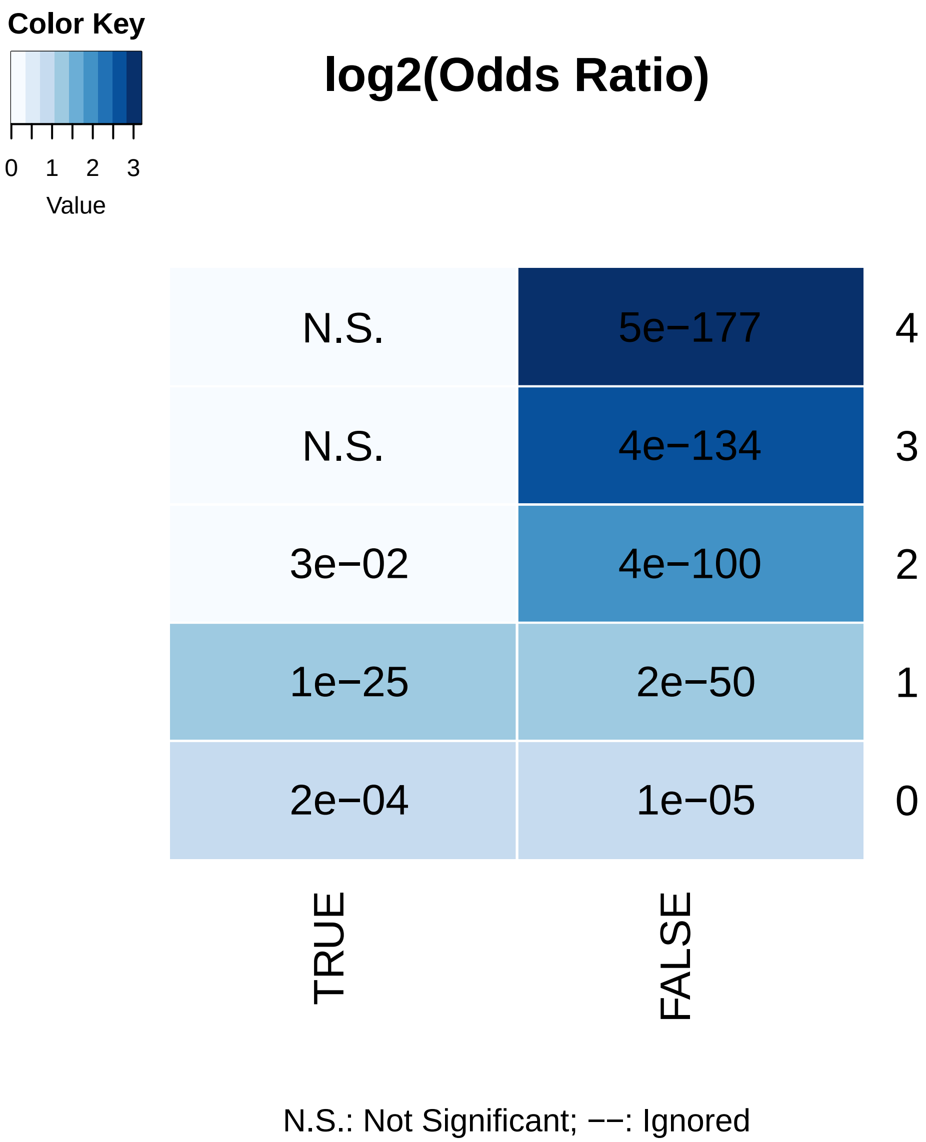

```
# Dher_tss |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Dher_tss_pval.tsv")
# Dher_tss |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
```

```
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Dher_tss_odds.ratio.tsv")
# TSS DesDud
Ddud_tss <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "DesDud") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_TSS_mark_list |>
                dplyr::filter(Sp == "DesDud") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Ddud_tss |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```




```
# Ddud_tss |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ddud_tss_pval.tsv")
# Ddud_tss |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ddud_tss_odds.ratio.tsv")
# TSS SchIsc
Sisc_tss <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list |>
                dplyr::filter(Sp == "SchIsc") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_TSS_mark_list |>
```

```
                dplyr::filter(Sp == "SchIsc") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Sisc_tss |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```

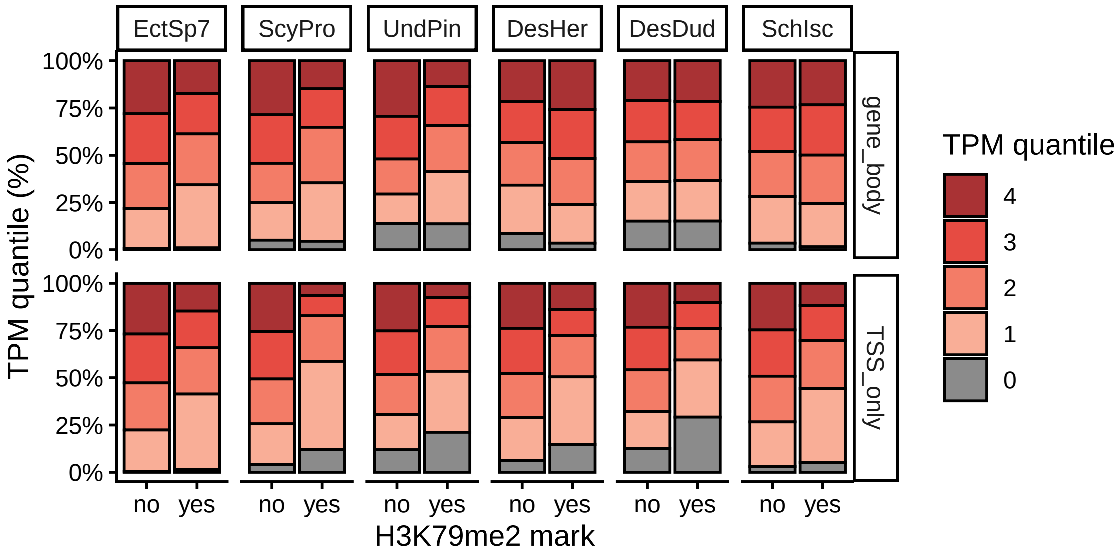



```
# Sisc_tss |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Sisc_tss_pval.tsv")
# Sisc_tss |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
```

```
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Sisc_tss_odds.ratio.tsv")
```

Another way to show the data but based on whether there is a mark at the TSS or over the whole gene (as x axis) and proportion of TPM quantiles (including TPM == 0 as 0th quantile) on the y axis.

```
all_k79.filt.TPM.filt.tss |>
        # dplyr::mutate(TPM_quantile = dplyr::case_when(
        #         TPM == 0 ~ "0",
        #         TPM > 0 ~ as.character(ntile(TPM, 4)),
        # ),
        # .by = Sp) |>
        # dplyr::mutate(Sp = factor(Sp, levels = c(
        #         "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
        # ))) |>
        dplyr::rename(gene_body = peak_K79, TSS_only = TSS_mark) |>
        tidyr::pivot_longer(cols = c(gene_body, TSS_only), names_to = "mark_type", values_to = "mark")
        dplyr::mutate(mark = case_when(
                mark == TRUE ~ "yes",
                .default = "no"
        )) |>
        ggplot2::ggplot(aes(x = factor(mark, levels = c("no","yes")), fill = factor(TPM_quantile, levels
        ggplot2::geom_bar(position = "fill", colour = "black") +
        ggplot2::scale_y_continuous(labels = scales::percent_format()) +
        ggplot2::labs(
                x = "H3K79me2 mark",
                y = "TPM quantile (%)",
                fill = "TPM quantile"
        ) +
        ggplot2::theme_classic() +
        # colour scale from grey for 0, beige 1 to red 4
        ggplot2::scale_fill_manual(values = rev(c("#8B8B8B", "#F9AE97", "#F37C67", "#E64B42", "#A93234")
#         ggplot2::scale_fill_brewer(palette = "Blues") +
        ggplot2::facet_grid(mark_type ~ Sp)
```

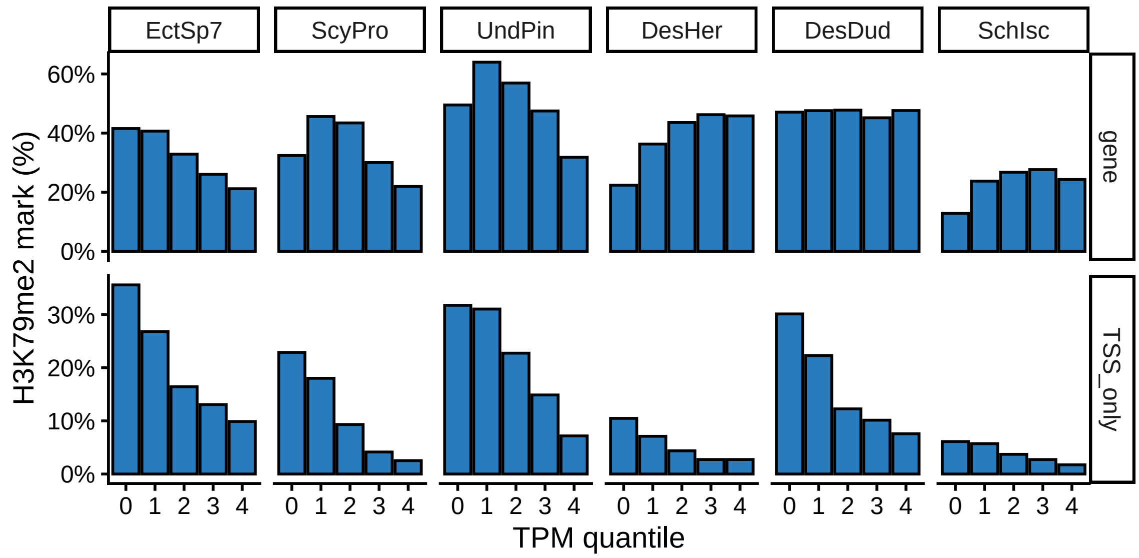


```r
# ggplot2::ggsave("figure/K79_tss_vs_gene_body_TPM_quantiles_barplot.pdf", width = 6, height = 3)

# statistical test for difference in TPM quantile distribution depending on mark presence
# use non-parametric test using information from TPM_quantile (which is a)
all_k79.filt.TPM.filt.tss |>
        dplyr::rename(gene_body = peak_K79, TSS_only = TSS_mark) |>
        tidyr::pivot_longer(cols = c(gene_body, TSS_only), names_to = "mark_type", values_to = "mark")
        dplyr::mutate(mark = case_when(
                mark == TRUE ~ "yes",
                .default = "no"
        )) |>
        dplyr::group_by(Sp, mark_type) |>
        rstatix::kruskal_test(TPM_quantile ~ mark) |>
        rstatix::adjust_pvalue(method = "bonferroni") |>
        dplyr::select(-c(3,4, 5), -df)
```

```
## # A tibble: 12 x 5
##    Sp     mark_type         p method             p.adj
##    <fct>  <chr>         <dbl> <chr>              <dbl>
##  1 EctSp7 TSS_only  1.35e-101 Kruskal-Wallis 1.62e-100
##  2 EctSp7 gene_body 4.98e- 94 Kruskal-Wallis 5.98e- 93
##  3 ScyPro TSS_only  5.97e-206 Kruskal-Wallis 7.16e-205
##  4 ScyPro gene_body 4.55e-127 Kruskal-Wallis 5.46e-126
##  5 UndPin TSS_only  9.44e-197 Kruskal-Wallis 1.13e-195
##  6 UndPin gene_body 5.13e-113 Kruskal-Wallis 6.16e-112
##  7 DesHer TSS_only  2.45e- 37 Kruskal-Wallis 2.94e- 36
##  8 DesHer gene_body 3.61e- 42 Kruskal-Wallis 4.33e- 41
##  9 DesDud TSS_only  6.78e-174 Kruskal-Wallis 8.14e-173
## 10 DesDud gene_body 5.66e-  1 Kruskal-Wallis 1   e+  0
## 11 SchIsc TSS_only  5.79e- 32 Kruskal-Wallis 6.95e- 31
## 12 SchIsc gene_body 1.92e-  3 Kruskal-Wallis 2.30e-  2
```

```r
# all_k79.filt.TPM.filt.tss |>
#         dplyr::mutate(Sp = factor(Sp, levels = c(
#                 "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
#         ))) |>
#         dplyr::rename(gene = peak_K79, TSS_only = TSS_mark) |>
#         tidyr::pivot_longer(!c(gene_id,TPM,Sp,TPM_quantile), names_to = "mark_type", values_to = "mar
#         dplyr::summarise(p_mark = mean(mark*1), .by = c(TPM_quantile, Sp,mark_type)) |>
#         ggplot2::ggplot(aes(x = TPM_quantile, y = p_mark, linetype = mark_type, group = mark_type)) +
#         ggplot2::geom_line(colour = "#297BBB") +
#         ggplot2::geom_point() +
#         ggplot2::theme_classic() +
#         # ggplot2::scale_colour_manual(values = c("grey80", "#297BBB")) +
#         ggplot2::scale_y_continuous(labels = scales::percent_format()) +
#         ggplot2::labs(
#                 x = "TPM quantile",
#                 y = "H3K79me2 mark (%)",
#                 linetype = "mark position"
#         ) +
#         ggplot2::facet_wrap(.~Sp, nrow = 1, scales = "free_y") +
#         ggplot2::theme(legend.position = "top")
#
# all_k79.filt.TPM.filt.tss |>
```

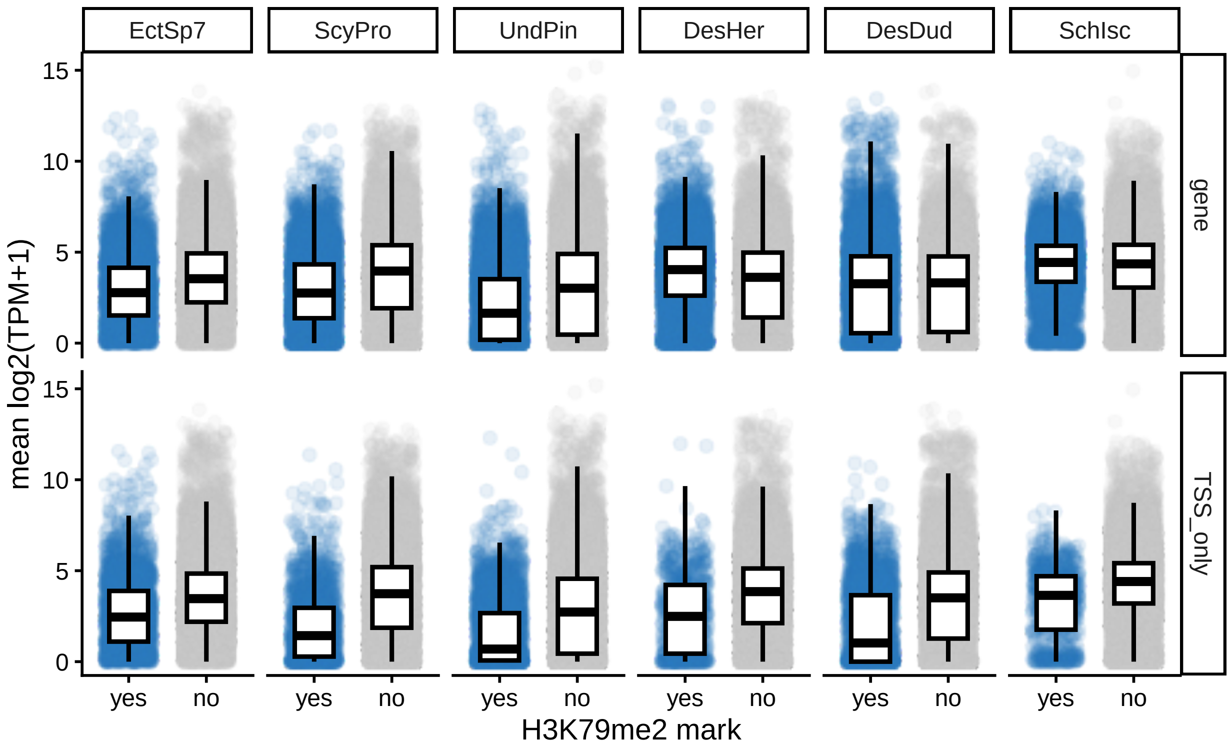
```r
#         dplyr::mutate(Sp = factor(Sp, levels = c(
#                 "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
#         ))) |>
#         dplyr::rename(gene = peak_K79, TSS_only = TSS_mark) |>
#         tidyr::pivot_longer(!c(gene_id,TPM,Sp,TPM_quantile), names_to = "mark_type", values_to = "mar
#         dplyr::summarise(p_mark = mean(mark*1), .by = c(TPM_quantile, Sp,mark_type)) |>
#         ggplot2::ggplot(aes(x = TPM_quantile, y = p_mark, group = mark_type)) +
#         ggplot2::geom_line(colour = "#297BBB") +
#         ggplot2::geom_point() +
#         ggplot2::theme_bw() +
#         ggplot2::scale_y_log10(labels = scales::percent_format()) +
#         ggplot2::annotation_logticks(sides = "l") +
#         # ggplot2::scale_y_continuous(labels = scales::percent_format()) +
#         ggplot2::labs(
#                 x = "TPM quantile",
#                 y = "H3K79me2 mark (%) (log-scaled)",
#                 linetype = "mark position"
#         ) +
#         ggplot2::facet_grid(mark_type ~ Sp, scales = "free_y") +
#         ggplot2::theme(legend.position = "top")

all_k79.filt.TPM.filt.tss |>
        dplyr::mutate(Sp = factor(Sp, levels = c(
                "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
        ))) |>
        dplyr::rename(gene = peak_K79, TSS_only = TSS_mark) |>
        tidyr::pivot_longer(!c(gene_id,TPM,Sp,TPM_quantile), names_to = "mark_type", values_to = "mark")
        dplyr::summarise(p_mark = mean(mark*1), .by = c(TPM_quantile, Sp,mark_type)) |>
        ggplot2::ggplot(aes(x = TPM_quantile, y = p_mark, group = mark_type)) +
        ggplot2::geom_col(fill = "#297BBB", colour = "black") +
        ggplot2::theme_classic() +
        ggplot2::scale_y_continuous(labels = scales::percent_format()) +
        ggplot2::labs(
                x = "TPM quantile",
                y = "H3K79me2 mark (%)",
                linetype = "mark position"
        ) +
        ggplot2::facet_grid(mark_type ~ Sp, scales = "free_y") +
        ggplot2::theme(legend.position = "top")
```




```
# boxplot style
all_k79.filt.tss <-
        Ec_k79.tss.f |> dplyr::mutate(Sp = "EctSp7") |>
        rbind(Spro_k79.tss.f |> dplyr::mutate(Sp = "ScyPro")) |>
        rbind(Und_k79.tss.f |> dplyr::mutate(Sp = "UndPin")) |>
        rbind(Dher_k79.tss.f |> dplyr::mutate(Sp = "DesHer")) |>
        dplyr::select(gene_id,
                      log2_TPM_plus_1 = log2_TPM_plus_1.Female,
                      peak_K79 = peak_K79.Female,
                      Sp, TSS_mark) |>
        rbind(Ddud_k79.tss |> dplyr::mutate(Sp = "DesDud") |> dplyr::select(1,3,4,6,7)) |>
        rbind(Si_k79.tss |> dplyr::mutate(Sp = "SchIsc") |> dplyr::select(1,3,4,6,7)) |>
        tidyr::drop_na()
```

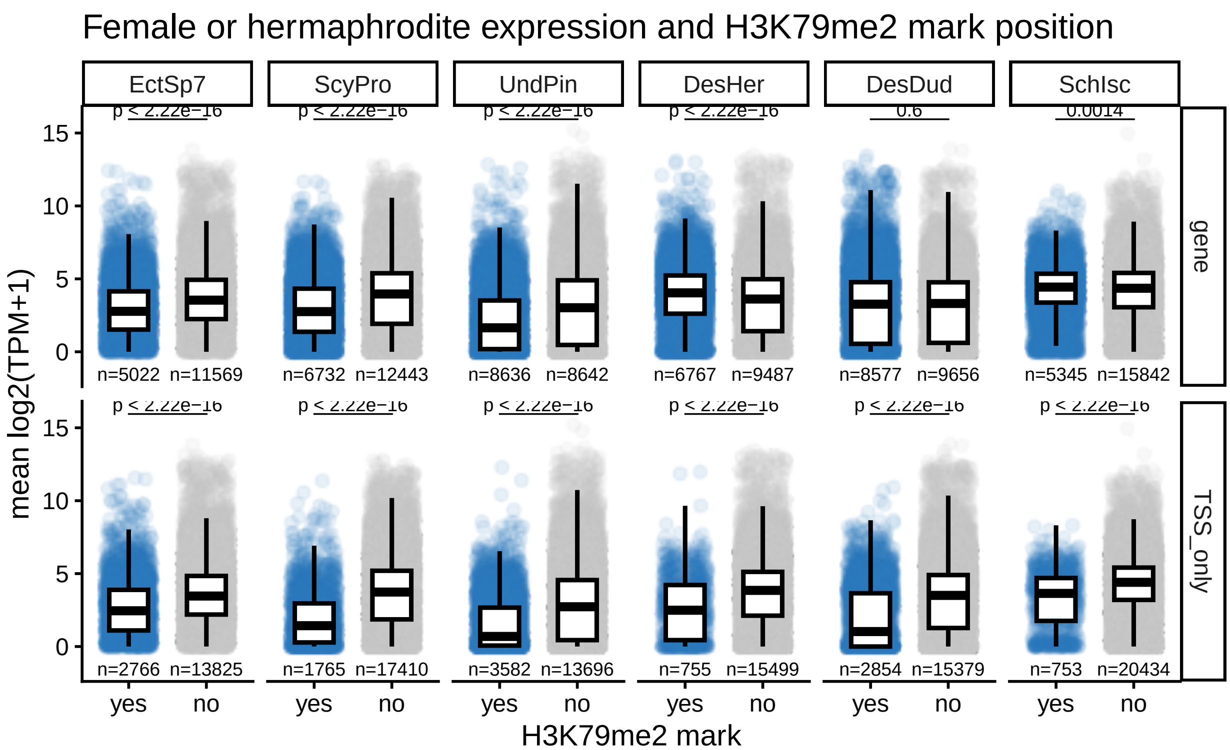
```
all_k79.filt.tss |>
        dplyr::mutate(Sp = factor(Sp, levels = c(
                "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
        ))) |>
        dplyr::mutate(peak_K79 = case_when(
                peak_K79 == TRUE ~ "yes",
                .default = "no"
        )) |>
        dplyr::mutate(TSS_mark = case_when(
                TSS_mark == TRUE ~ "yes",
                .default = "no"
        )) |>
        dplyr::rename(gene = peak_K79, TSS_only = TSS_mark) |>
        tidyr::pivot_longer(!c(gene_id,log2_TPM_plus_1,Sp), names_to = "mark_type", values_to = "mark")
        dplyr::mutate(mark = factor(mark, levels = c("yes", "no"))) |>
        ggplot2::ggplot(aes(x = mark, y = log2_TPM_plus_1, colour = mark)) |>
        plot_style2() +
        ggplot2::theme_classic() +
        ggplot2::labs(
                x = "H3K79me2 mark",
                y = "mean log2(TPM+1)"
        ) +
```

```
        ggplot2::facet_grid(mark_type ~ Sp) +
        ggplot2::theme(legend.position = "")
```




```
all_k79.filt.tss |>
        dplyr::mutate(Sp = factor(Sp, levels = c(
                "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
        ))) |>
        dplyr::mutate(peak_K79 = case_when(
                peak_K79 == TRUE ~ "yes",
                .default = "no"
        )) |>
        dplyr::mutate(TSS_mark = case_when(
                TSS_mark == TRUE ~ "yes",
                .default = "no"
        )) |>
        dplyr::rename(gene = peak_K79, TSS_only = TSS_mark) |>
        tidyr::pivot_longer(!c(gene_id,log2_TPM_plus_1,Sp), names_to = "mark_type", values_to = "mark")
        dplyr::mutate(mark = factor(mark, levels = c("yes", "no"))) |>
        ggplot2::ggplot(aes(x = mark, y = log2_TPM_plus_1, colour = mark)) |>
        plot_style2() +
        ggplot2::theme_classic() +
        ggplot2::labs(
                x = "H3K79me2 mark",
                y = "mean log2(TPM+1)"
        ) +
        ggplot2::facet_grid(mark_type ~ Sp) +
        ggplot2::theme(legend.position = "") +
        ggpubr::stat_compare_means(
                label = "p.adj",
                method = "wilcox.test",
```

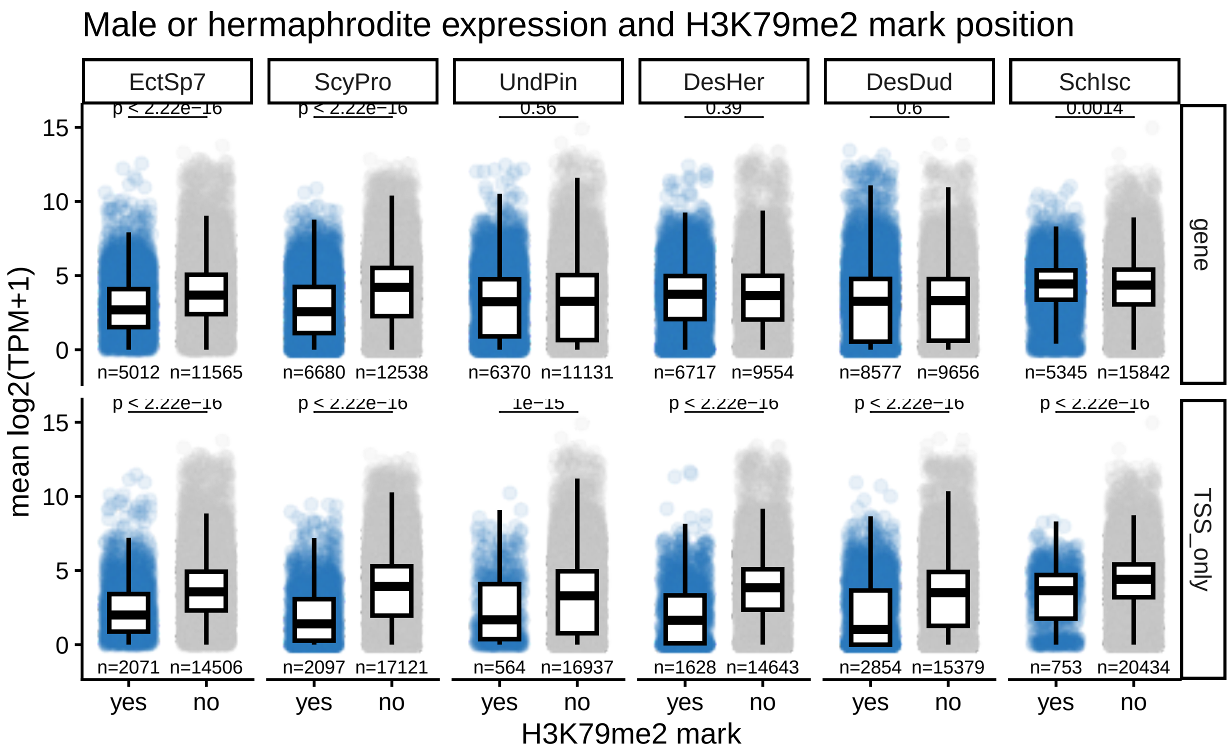
```
            comparisons = list(c("yes", "no")),
            p.adjust.method = "bonferroni",
            tip.length = 0,
            size = 2.5
        ) +
        EnvStats::stat_n_text(size = 2.5) +
        ggplot2::labs(title = "Female or hermaphrodite expression and H3K79me2 mark position")
```




```
# boxplot style
all_k79.filt.tss.m <-
        Ec_k79.tss.m |> dplyr::mutate(Sp = "EctSp7") |>
        rbind(Spro_k79.tss.m |> dplyr::mutate(Sp = "ScyPro")) |>
        rbind(Und_k79.tss.m |> dplyr::mutate(Sp = "UndPin")) |>
        rbind(Dher_k79.tss.m |> dplyr::mutate(Sp = "DesHer")) |>
        dplyr::select(gene_id,
                      log2_TPM_plus_1 = log2_TPM_plus_1.Male,
                      peak_K79 = peak_K79.Male,
                      Sp, TSS_mark) |>
        rbind(Ddud_k79.tss |> dplyr::mutate(Sp = "DesDud") |> dplyr::select(1,3,4,6,7)) |>
        rbind(Si_k79.tss |> dplyr::mutate(Sp = "SchIsc") |> dplyr::select(1,3,4,6,7)) |>
        tidyr::drop_na()
```

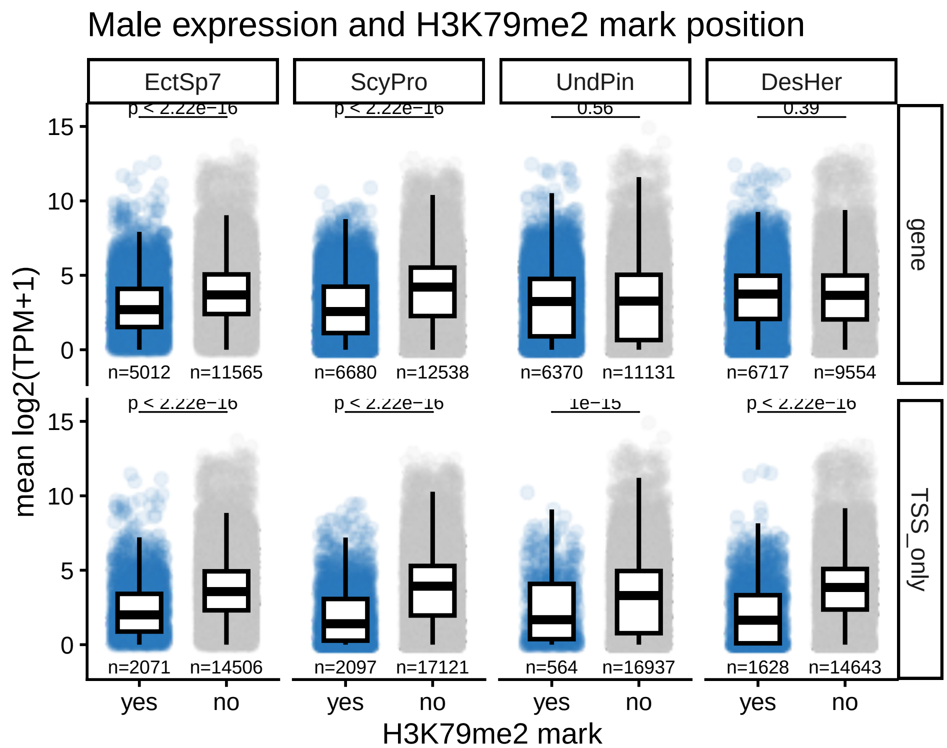
```
# all_k79.filt.tss.m |>
#         dplyr::mutate(Sp = factor(Sp, levels = c(
#                 "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
#         ))) |>
#         dplyr::mutate(peak_K79 = case_when(
#                 peak_K79 == TRUE ~ "yes",
#                 .default = "no"
#         )) |>
```

```
#         dplyr::mutate(TSS_mark = case_when(
#                 TSS_mark == TRUE ~ "yes",
#                 .default = "no"
#         )) |>
#         dplyr::rename(gene = peak_K79, TSS_only = TSS_mark) |>
#         tidyr::pivot_longer(!c(gene_id,log2_TPM_plus_1,Sp), names_to = "mark_type", values_to = "mark
#         dplyr::mutate(mark = factor(mark, levels = c("yes", "no"))) |>
#         ggplot2::ggplot(aes(x = mark, y = log2_TPM_plus_1, colour = mark)) |>
#         plot_style2() +
#         ggplot2::theme_classic() +
#         ggplot2::labs(
#                 x = "H3K79me2 mark",
#                 y = "mean log2(TPM+1)"
#         ) +
#         ggplot2::facet_grid(mark_type ~ Sp) +
#         ggplot2::theme(legend.position = "")

all_k79.filt.tss.m |>
        dplyr::mutate(Sp = factor(Sp, levels = c(
                "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
        ))) |>
        dplyr::mutate(peak_K79 = case_when(
                peak_K79 == TRUE ~ "yes",
                .default = "no"
        )) |>
        dplyr::mutate(TSS_mark = case_when(
                TSS_mark == TRUE ~ "yes",
                .default = "no"
        )) |>
        dplyr::rename(gene = peak_K79, TSS_only = TSS_mark) |>
        tidyr::pivot_longer(!c(gene_id,log2_TPM_plus_1,Sp), names_to = "mark_type", values_to = "mark")
        dplyr::mutate(mark = factor(mark, levels = c("yes", "no"))) |>
        ggplot2::ggplot(aes(x = mark, y = log2_TPM_plus_1, colour = mark)) |>
        plot_style2() +
        ggplot2::theme_classic() +
        ggplot2::labs(
                x = "H3K79me2 mark",
                y = "mean log2(TPM+1)"
        ) +
        ggplot2::facet_grid(mark_type ~ Sp) +
        ggplot2::theme(legend.position = "") +
        ggpubr::stat_compare_means(
                label = "p.adj",
                method = "wilcox.test",
                comparisons = list(c("yes", "no")),
                p.adjust.method = "bonferroni",
                tip.length = 0,
                size = 2.5
        ) +
        EnvStats::stat_n_text(size = 2.5) +
        ggplot2::labs(title = "Male or hermaphrodite expression and H3K79me2 mark position")
```




```r
# all_k79.filt.tss.m |>
#         dplyr::mutate(Sp = factor(Sp, levels = c(
#                 "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
#         ))) |>
#         dplyr::mutate(peak_K79 = case_when(
#                 peak_K79 == TRUE ~ "yes",
#                 .default = "no"
#         )) |>
#         dplyr::mutate(TSS_mark = case_when(
#                 TSS_mark == TRUE ~ "yes",
#                 .default = "no"
#         )) |>
#         dplyr::rename(gene = peak_K79, TSS_only = TSS_mark) |>
#         tidyr::pivot_longer(!c(gene_id,log2_TPM_plus_1,Sp), names_to = "mark_type", values_to = "mark
#         dplyr::mutate(mark = factor(mark, levels = c("yes", "no"))) |>
#         ggplot2::ggplot(aes(x = mark, y = log2_TPM_plus_1, colour = mark)) |>
#         plot_style2() +
#         ggplot2::theme_classic() +
#         ggplot2::labs(
#                 x = "H3K79me2 mark",
#                 y = "mean log2(TPM+1)"
#         ) +
#         ggplot2::facet_grid(mark_type ~ Sp) +
#         ggplot2::theme(legend.position = "")

all_k79.filt.tss.m |>
        dplyr::mutate(Sp = factor(Sp, levels = c(
                "EctSp7", "ScyPro", "UndPin", "DesHer", "DesDud", "SchIsc"
        ))) |>
        dplyr::filter(Sp %in% c("EctSp7", "ScyPro", "UndPin", "DesHer")) |>
```

```r
    dplyr::mutate(peak_K79 = case_when(
            peak_K79 == TRUE ~ "yes",
            .default = "no"
    )) |>
    dplyr::mutate(TSS_mark = case_when(
            TSS_mark == TRUE ~ "yes",
            .default = "no"
    )) |>
    dplyr::rename(gene = peak_K79, TSS_only = TSS_mark) |>
    tidyr::pivot_longer(!c(gene_id,log2_TPM_plus_1,Sp), names_to = "mark_type", values_to = "mark")
    dplyr::mutate(mark = factor(mark, levels = c("yes", "no"))) |>
    ggplot2::ggplot(aes(x = mark, y = log2_TPM_plus_1, colour = mark)) |>
    plot_style2() +
    ggplot2::theme_classic() +
    ggplot2::labs(
            x = "H3K79me2 mark",
            y = "mean log2(TPM+1)"
    ) +
    ggplot2::facet_grid(mark_type ~ Sp) +
    ggplot2::theme(legend.position = "") +
    ggpubr::stat_compare_means(
            label = "p.adj",
            method = "wilcox.test",
            comparisons = list(c("yes", "no")),
            p.adjust.method = "bonferroni",
            tip.length = 0,
            size = 2.5
    ) +
    EnvStats::stat_n_text(size = 2.5) +
    ggplot2::labs(title = "Male expression and H3K79me2 mark position")
```



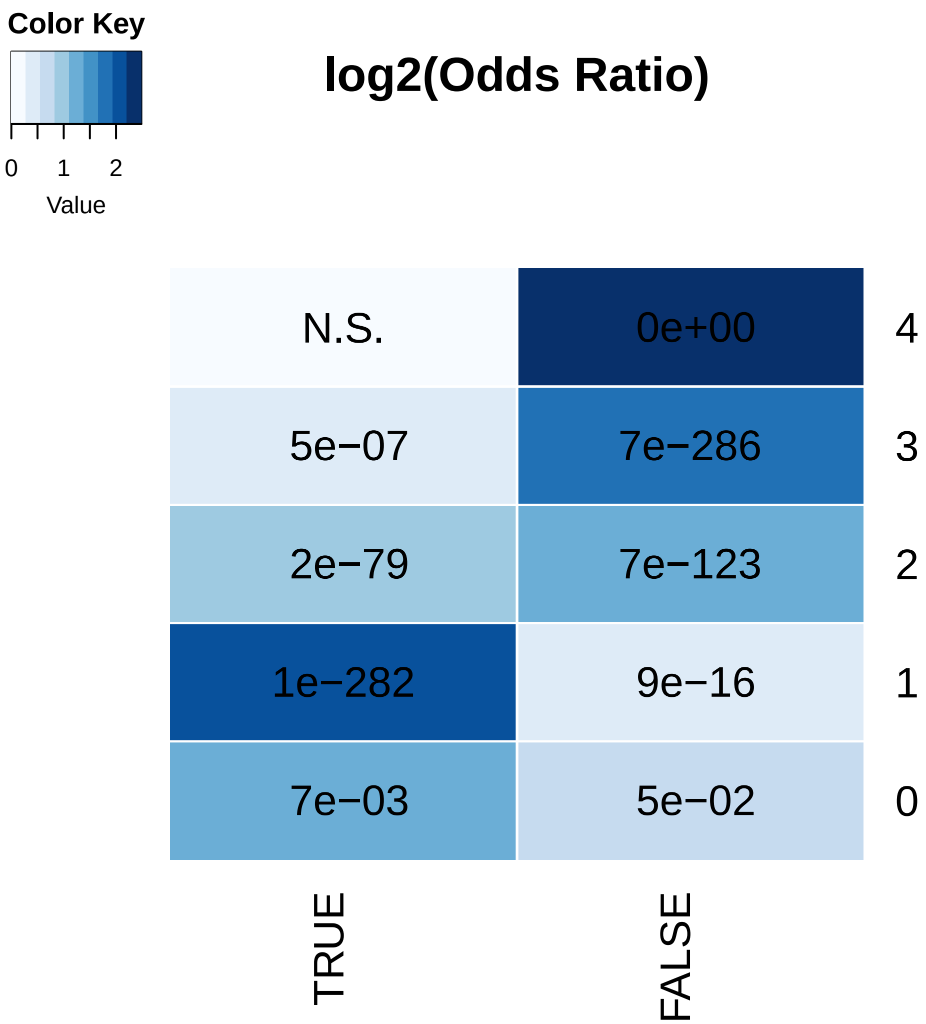
Focus instead on males.

```
# full data
all_k79.filt.TPM.tss.m <-
        Ec_k79.tss.m |> dplyr::mutate(Sp = "EctSp7") |>
        rbind(Spro_k79.tss.m |> dplyr::mutate(Sp = "ScyPro")) |>
        rbind(Und_k79.tss.m |> dplyr::mutate(Sp = "UndPin")) |>
        rbind(Dher_k79.tss.m |> dplyr::mutate(Sp = "DesHer")) |>
        dplyr::select(gene_id,
                      TPM = TPM.Male,
                      peak_K79 = peak_K79.Male,
                      Sp, TSS_mark) |>
        tidyr::drop_na()

TPM_threshold = 0
all_k79.filt.TPM.filt.tss.m <-
        all_k79.filt.TPM.tss.m |>
        dplyr::group_by(Sp) |>
        dplyr::mutate(
                quant_tmp = {
                        x = TPM
                        x[TPM > TPM_threshold] = ntile(x[TPM > TPM_threshold], 4)
                        x[TPM <= TPM_threshold] = NA_integer_
                        x
                },
                TPM_quantile = dplyr::case_when(
                        TPM <= TPM_threshold ~ "0",
                        TRUE ~ as.character(quant_tmp)
                )
        ) |>
```

```
        dplyr::select(-quant_tmp) |>
        dplyr::ungroup() |>
        dplyr::mutate(Sp = factor(Sp, levels = c(
                "EctSp7", "ScyPro", "UndPin", "DesHer" #, "DesDud", "SchIsc"
        )))
```

Plot this data as a heatmap with percentage of genes with mark in each TPM quantile.

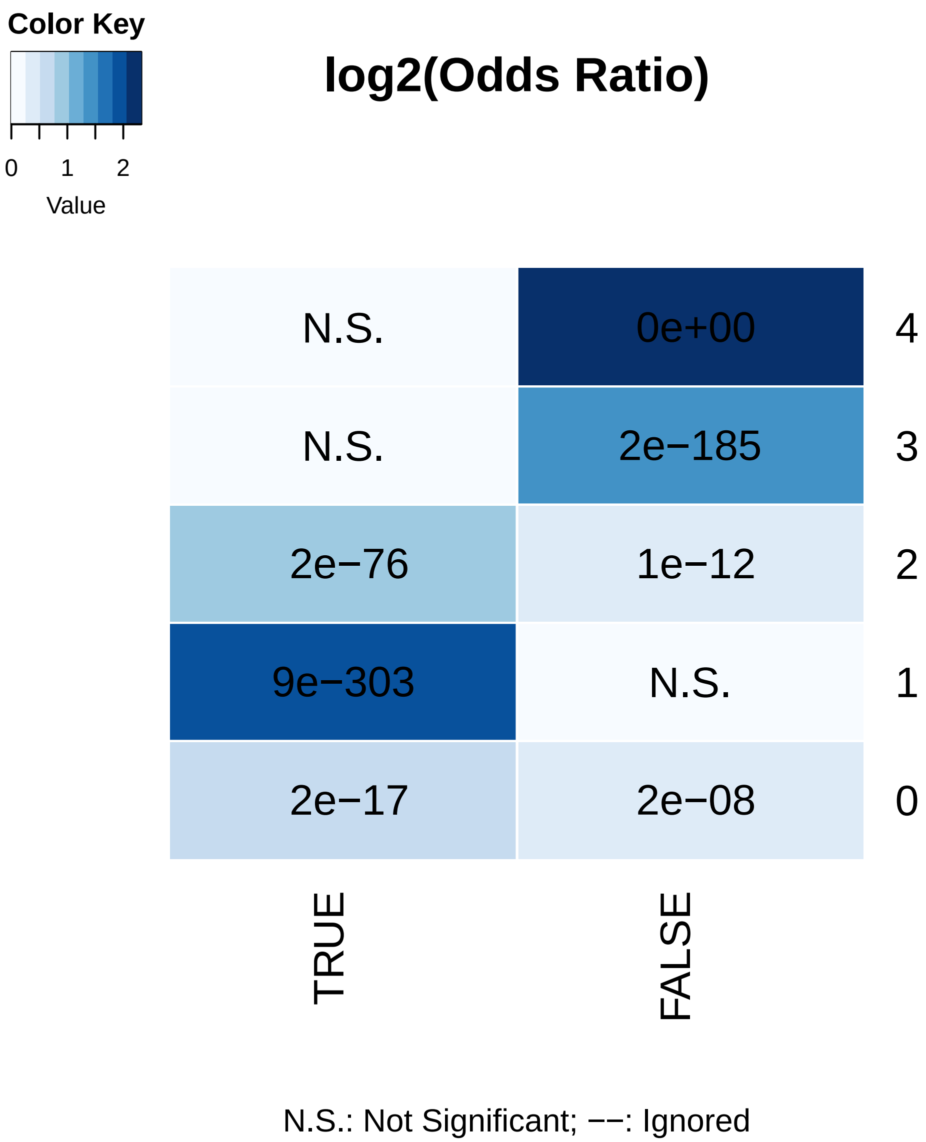
```
# using the GeneOverlap package to test for significance of overlap between genes with TSS mark and gen
library(GeneOverlap)
all_k79.filt.TPM.filt.tss_TPM_quantile_list.m <-
        all_k79.filt.TPM.filt.tss.m |>
        summarise(gene_id = list(gene_id), .by = c(TPM_quantile, Sp)) |>
        dplyr::mutate(TPM_quantile = factor(TPM_quantile, levels = rev(c("0","1","2","3","4")))) |>
        dplyr::arrange(TPM_quantile)

all_k79.filt.TPM.filt.tss_peak79_list.m <-
        all_k79.filt.TPM.filt.tss.m |>
        summarise(gene_id = list(gene_id), .by = c(peak_K79, Sp)) |>
        dplyr::mutate(peak_K79 = factor(peak_K79, levels = c(TRUE, FALSE))) |>
        dplyr::arrange(peak_K79)

# gene body EctSp7
Ec_body.m <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list.m |>
                dplyr::filter(Sp == "EctSp7") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_peak79_list.m |>
                dplyr::filter(Sp == "EctSp7") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Ec_body.m |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



N.S.: Not Significant; −−: Ignored

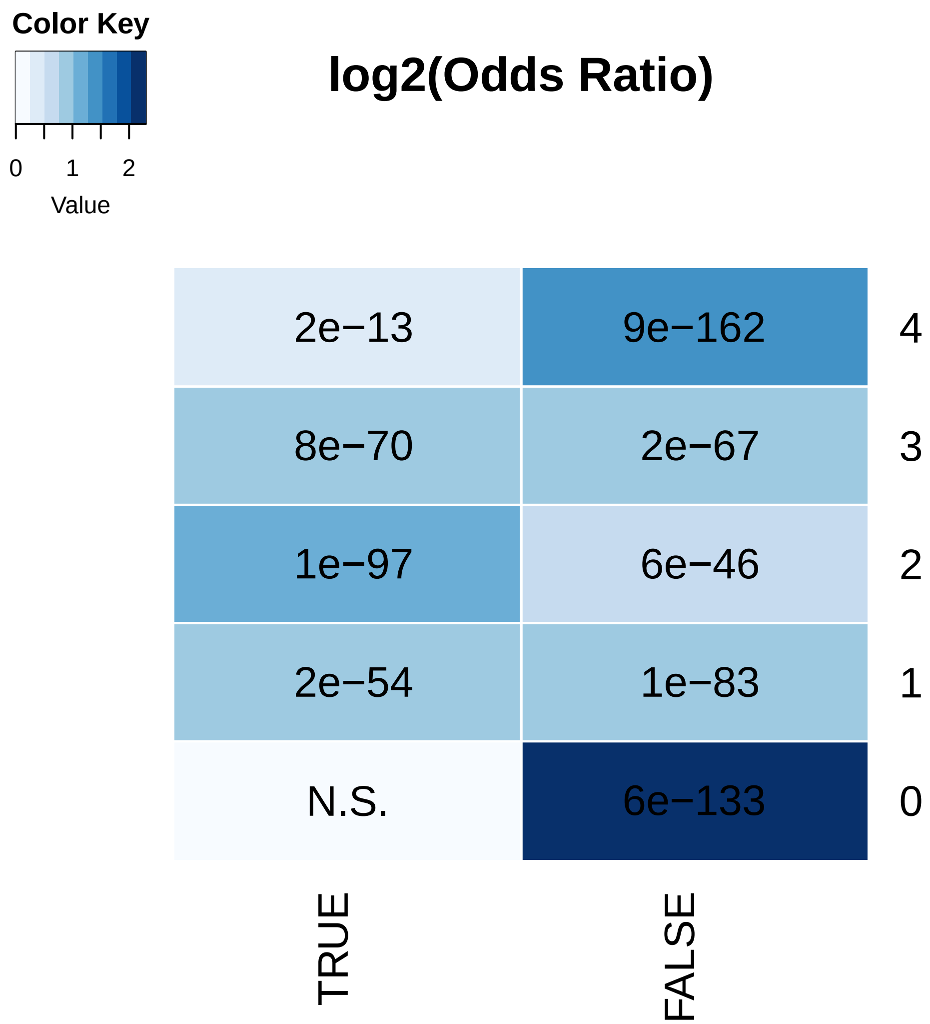
```
# Ec_body.m |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ec_body_m_pval.tsv")
# Ec_body.m |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ec_body_m_odds.ratio.tsv")

# gene body Spro
Spro_body.m <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list.m |>
                dplyr::filter(Sp == "ScyPro") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
```

```
        all_k79.filt.TPM.filt.tss_peak79_list.m |>
                dplyr::filter(Sp == "ScyPro") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Spro_body.m |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```




```
# Spro_body.m |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Spro_body_m_pval.tsv")
# Spro_body.m |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
```

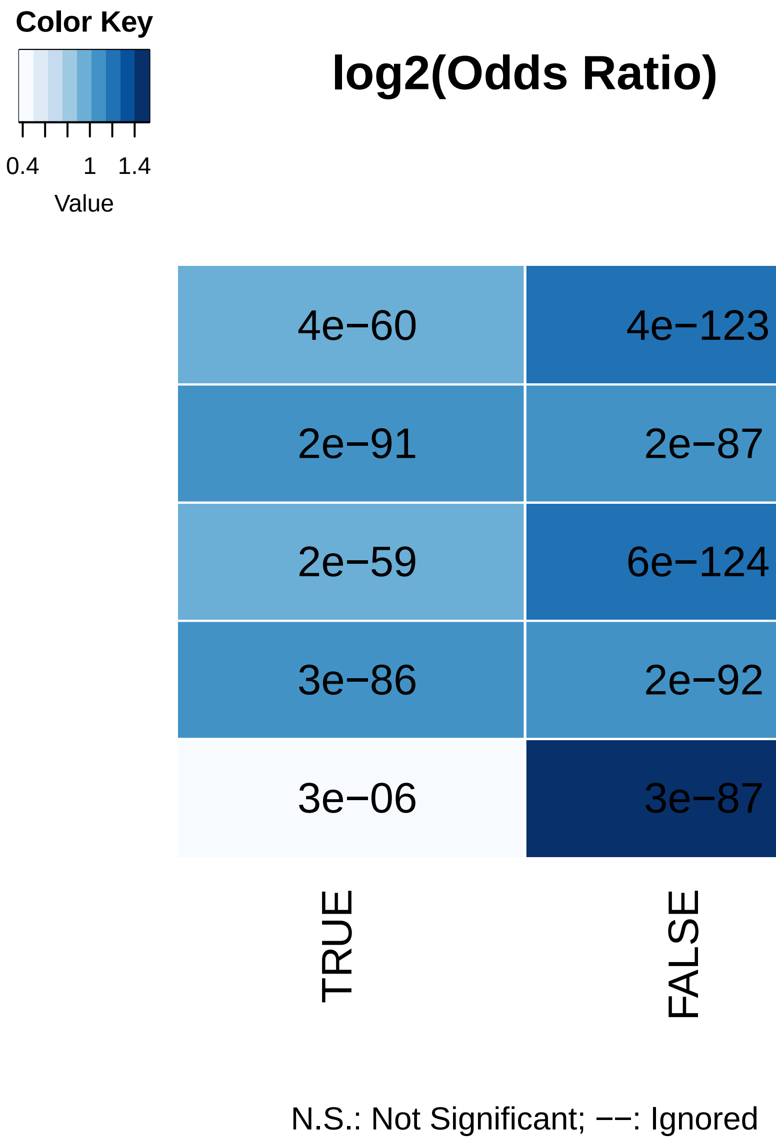
```
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Spro_body_m_odds.ratio.tsv")

# gene body UndPin
Upin_body.m <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list.m |>
                dplyr::filter(Sp == "UndPin") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_peak79_list.m |>
                dplyr::filter(Sp == "UndPin") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Upin_body.m |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



N.S.: Not Significant; −−: Ignored

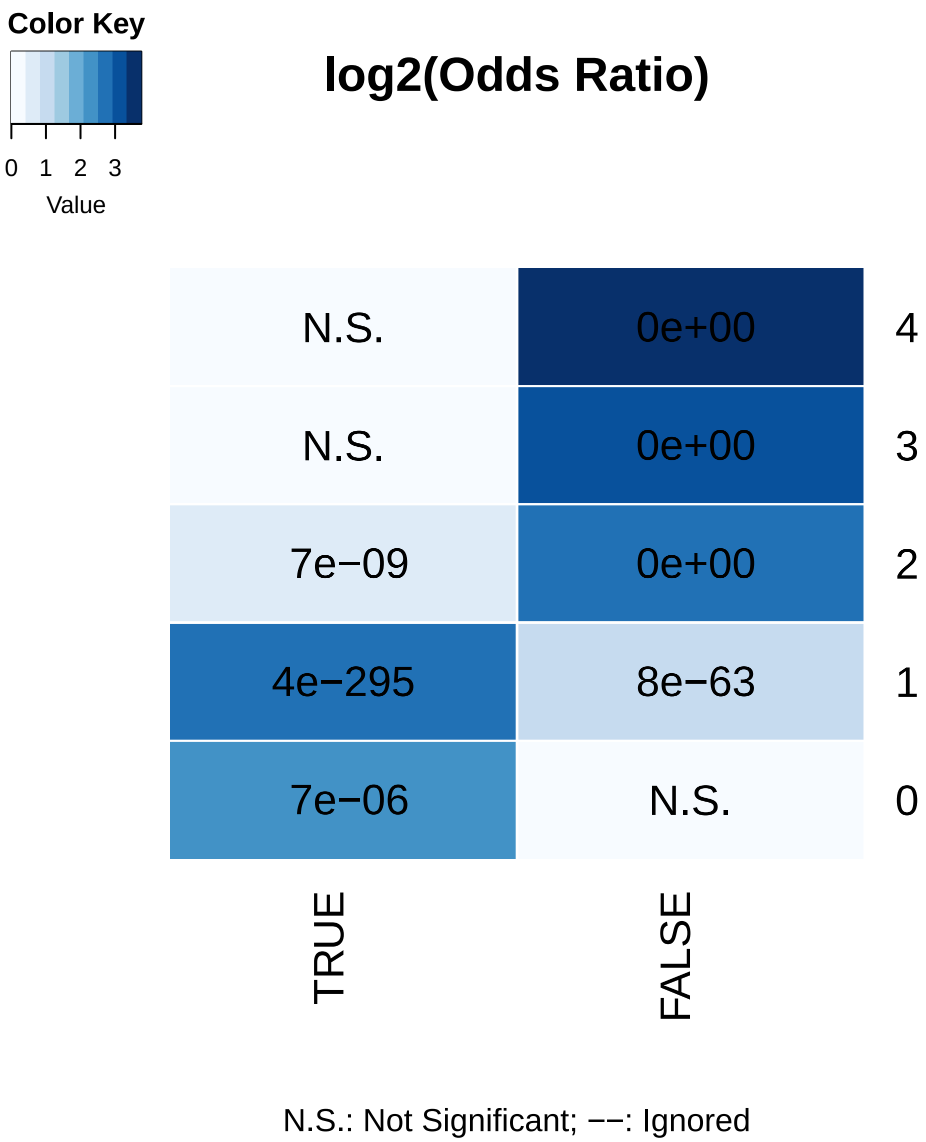
```r
# Upin_body.m |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Upin_body_m_pval.tsv")
# Upin_body.m |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Upin_body_m_odds.ratio.tsv")

# gene body DesHer
Dher_body.m <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list.m |>
                dplyr::filter(Sp == "DesHer") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
```

```
        all_k79.filt.TPM.filt.tss_peak79_list.m |>
                dplyr::filter(Sp == "DesHer") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Dher_body.m |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



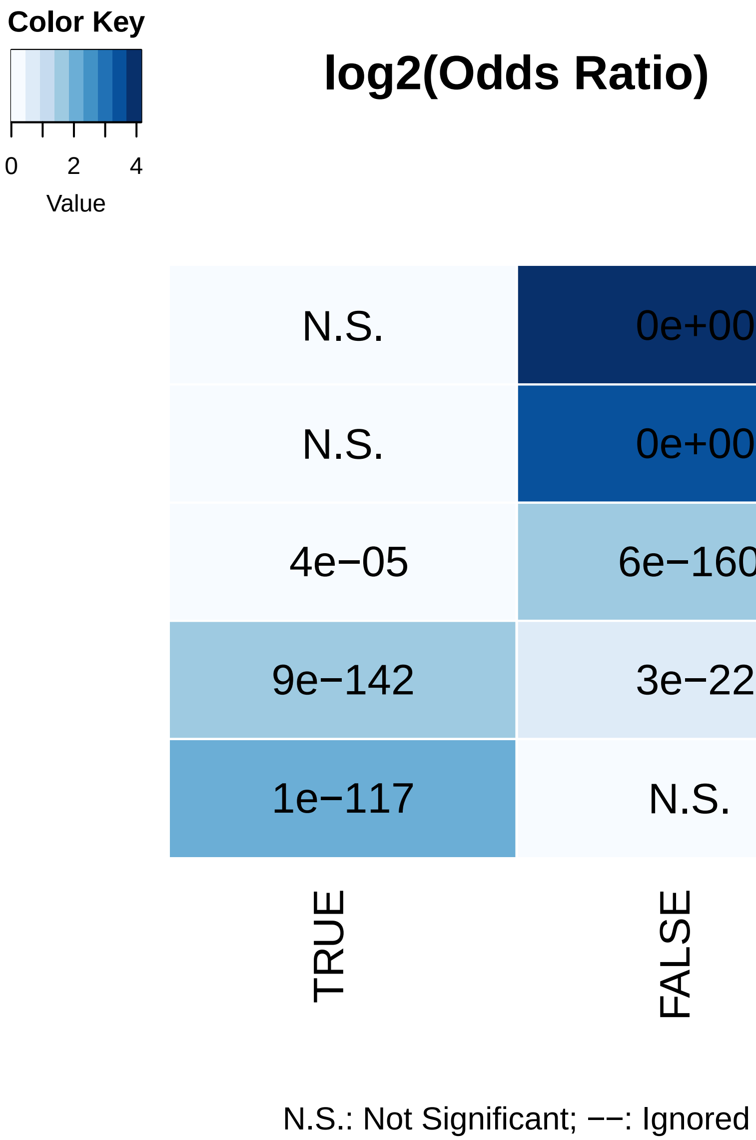

```
# Dher_body.m |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Dher_body_m_pval.tsv")
# Dher_body.m |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
```

```
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Dher_body_m_odds.ratio.tsv")


all_k79.filt.TPM.filt.tss_TSS_mark_list.m <-
        all_k79.filt.TPM.filt.tss.m |>
        summarise(gene_id = list(gene_id), .by = c(TSS_mark, Sp)) |>
        dplyr::mutate(TSS_mark = factor(TSS_mark, levels = c(TRUE, FALSE))) |>
        dplyr::arrange(TSS_mark)

# TSS EctSp7
Ec_TSS.m <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list.m |>
                dplyr::filter(Sp == "EctSp7") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_TSS_mark_list.m |>
                dplyr::filter(Sp == "EctSp7") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Ec_TSS.m |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



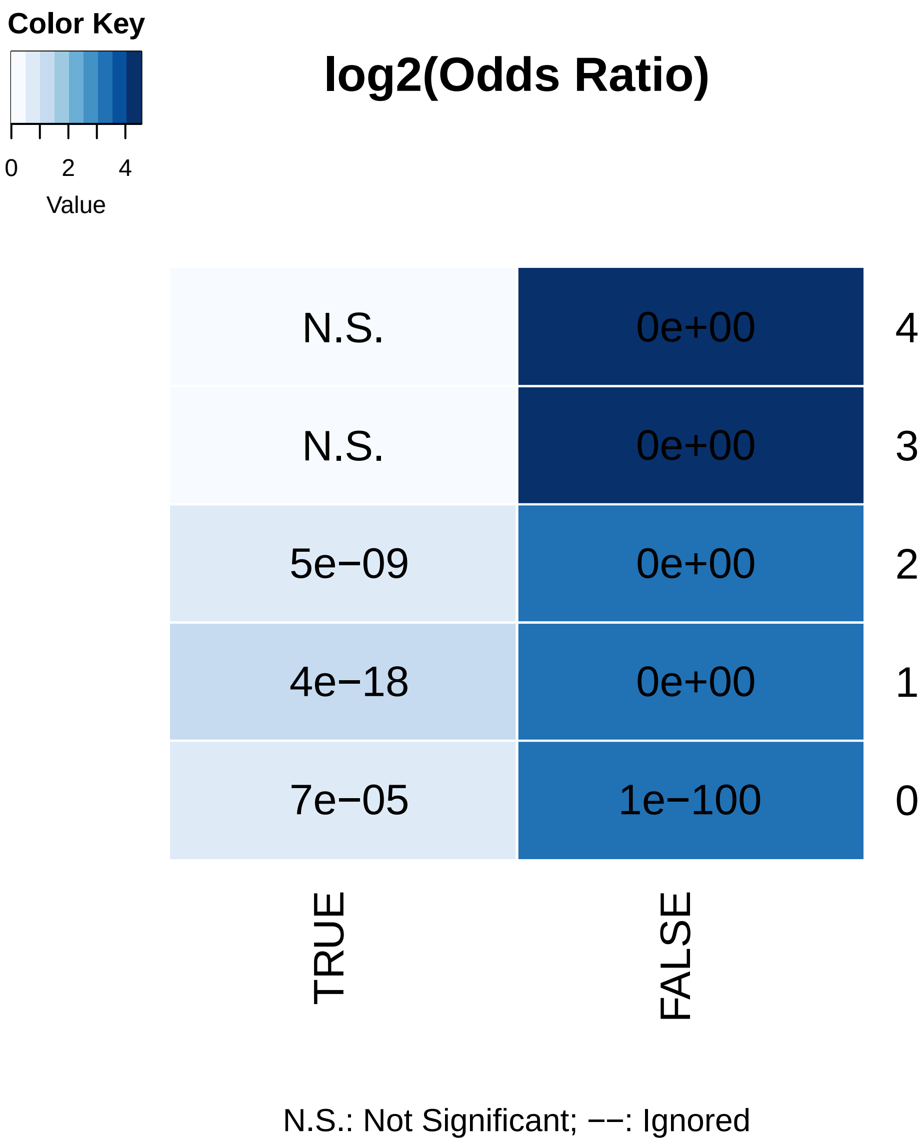

```
# Ec_TSS.m |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ec_TSS_m_pval.tsv")
# Ec_TSS.m |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Ec_TSS_m_odds.ratio.tsv")

# TSS Spro
Spro_TSS.m <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list.m |>
                dplyr::filter(Sp == "ScyPro") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
```

```r
        all_k79.filt.TPM.filt.tss_TSS_mark_list.m |>
                dplyr::filter(Sp == "ScyPro") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Spro_TSS.m |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```



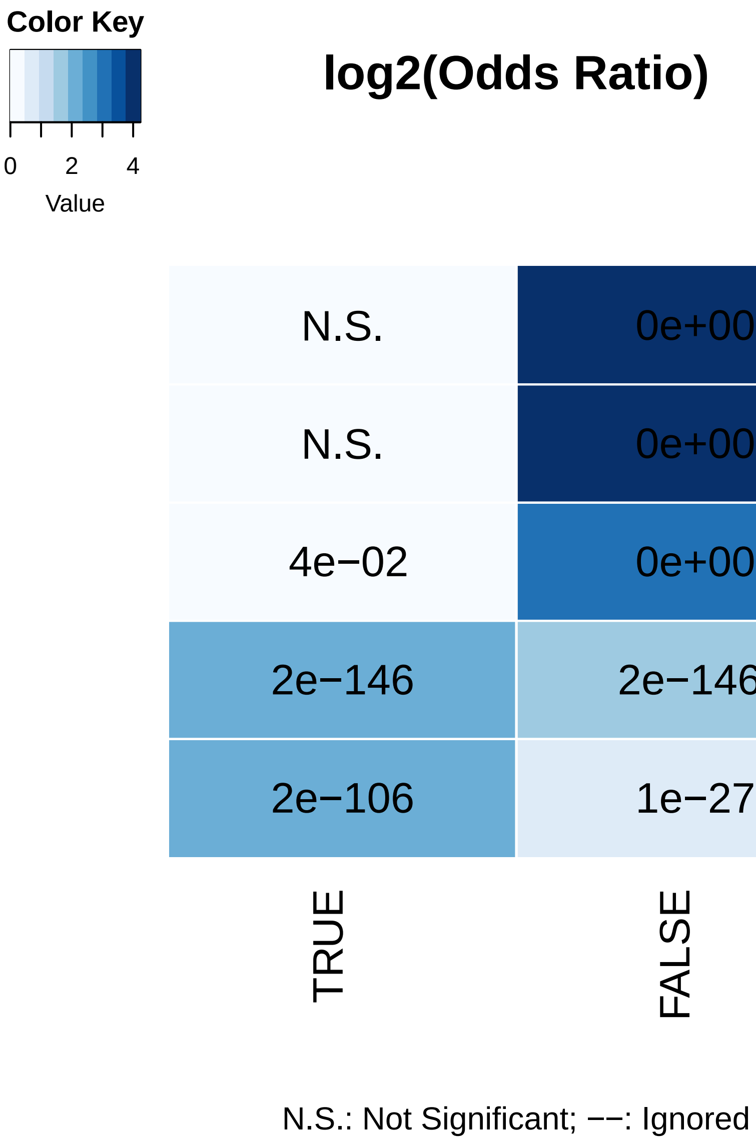

```r
# Spro_TSS.m |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Spro_TSS_m_pval.tsv")
# Spro_TSS.m |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
```

```r
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Spro_TSS_m_odds.ratio.tsv")

# TSS UndPin
Upin_TSS.m <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list.m |>
                dplyr::filter(Sp == "UndPin") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
        all_k79.filt.TPM.filt.tss_TSS_mark_list.m |>
                dplyr::filter(Sp == "UndPin") |>
                dplyr::select(-Sp) |>
                tibble::deframe())

Upin_TSS.m |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```




```
# Upin_TSS.m |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Upin_TSS_m_pval.tsv")
# Upin_TSS.m |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Upin_TSS_m_odds.ratio.tsv")

# TSS DesHer
Dher_TSS.m <-
        GeneOverlap::newGOM(
        all_k79.filt.TPM.filt.tss_TPM_quantile_list.m |>
                dplyr::filter(Sp == "DesHer") |>
                dplyr::select(-Sp) |>
                tibble::deframe(),
```

```r
        all_k79.filt.TPM.filt.tss_TSS_mark_list.m |>
                dplyr::filter(Sp == "DesHer") |>
                dplyr::select(-Sp) |>
                tibble::deframe())
Dher_TSS.m |>
        GeneOverlap::drawHeatmap(log.scale=T,
                                 grid.col="Blues",
                                 note.col="black",
                                 adj.p = T)
```




```r
# Dher_TSS.m |>
#         GeneOverlap::getMatrix(name = "pval") |>
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Dher_TSS_m_pval.tsv")
# Dher_TSS.m |>
#         GeneOverlap::getMatrix(name = "odds.ratio") |>
```

```r
#         tidyr::as_tibble() |>
#         tibble::rownames_to_column("TPM_quantile") |>
#         readr::write_tsv("table/Dher_TSS_m_odds.ratio.tsv")
```

```r
devtools::session_info(pkgs = "attached")
```

```
## - Session info ------------------------------------------------------------------
##  setting  value
##  version  R version 4.4.0 (2024-04-24)
##  os       macOS 15.7.2
##  system   x86_64, darwin20
##  ui       X11
##  language (EN)
##  collate  en_US.UTF-8
##  ctype    en_US.UTF-8
##  tz       Asia/Bangkok
##  date     2025-12-04
##  pandoc   3.4 @ /Applications/RStudio.app/Contents/Resources/app/quarto/bin/tools/x86_64/ (via rmark
##
## - Packages ----------------------------------------------------------------------
##  package       * version date (UTC) lib source
##  BiocGenerics  * 0.50.0  2024-04-30 [1] Bioconductor 3.19 (R 4.4.0)
##  ChIPpeakAnno  * 3.38.0  2024-04-30 [1] Bioconductor 3.19 (R 4.4.0)
##  dplyr         * 1.1.4   2023-11-17 [1] CRAN (R 4.4.0)
##  forcats       * 1.0.0   2023-01-29 [1] CRAN (R 4.4.0)
##  GeneOverlap   * 1.40.0  2024-04-30 [1] Bioconductor 3.19 (R 4.4.0)
##  GenomeInfoDb  * 1.40.1  2024-05-24 [1] Bioconductor 3.19 (R 4.4.0)
##  GenomicRanges * 1.56.2  2024-10-09 [1] Bioconductor 3.19 (R 4.4.1)
##  ggplot2       * 4.0.0   2025-09-11 [1] CRAN (R 4.4.1)
##  ggrastr       * 1.0.2   2023-06-01 [1] CRAN (R 4.4.0)
##  IRanges       * 2.38.1  2024-07-03 [1] Bioconductor 3.19 (R 4.4.1)
##  lubridate     * 1.9.4   2024-12-08 [1] CRAN (R 4.4.1)
##  purrr         * 1.1.0   2025-07-10 [1] CRAN (R 4.4.1)
##  readr         * 2.1.5   2024-01-10 [1] CRAN (R 4.4.0)
##  readxl        * 1.4.3   2023-07-06 [1] CRAN (R 4.4.0)
##  reshape2      * 1.4.4   2020-04-09 [1] CRAN (R 4.4.0)
##  rtracklayer   * 1.64.0  2024-05-01 [1] Bioconductor 3.19 (R 4.4.0)
##  S4Vectors     * 0.42.1  2024-07-03 [1] Bioconductor 3.19 (R 4.4.1)
##  scales        * 1.4.0   2025-04-24 [1] CRAN (R 4.4.1)
##  stringr       * 1.5.2   2025-09-08 [1] CRAN (R 4.4.1)
##  tibble        * 3.3.0   2025-06-08 [1] CRAN (R 4.4.1)
##  tidyr         * 1.3.1   2024-01-24 [1] CRAN (R 4.4.0)
##  tidyverse     * 2.0.0   2023-02-22 [1] CRAN (R 4.4.0)
##
##  [1] /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library
##
## ---------------------------------------------------------------------------------
```